
Software Requirements and Design Document

for

INDICIUM

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SCOPE

A Secure Digital Evidence Management System

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1. Introduction

1.1. Purpose

Indicium is a secure digital evidence management system designed to support forensic investigations and maintain the integrity of digital evidence throughout its lifecycle. The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements, system behavior, architecture, and design considerations of the Indicium platform.

This document serves as a reference for developers, testers, instructors, and stakeholders to understand the objectives and expected behavior of the system. It also establishes a structured guideline for implementation, testing, and future maintenance of the project.

1.2. Project Scope

Indicium is a desktop-based forensic evidence management system developed using Java. The system is intended to help investigators securely collect, organize, store, retrieve, and analyze digital evidence while preserving the chain of custody.

The platform integrates cryptographic hashing, role-based access control, audit logging, evidence linking, and timeline reconstruction to ensure that all digital evidence remains authentic and traceable. The system focuses on secure local evidence management rather than advanced forensic analysis or external law enforcement integration.

1.3. Title

Indicium – A Secure Digital Evidence Management System

1.4. Objectives

- Develop a secure evidence management system for digital forensic investigations.
- Enable investigators to securely ingest, organize, and review digital evidence linked to investigation cases.
- Implement cryptographic hashing to verify evidence integrity and detect tampering.
- Provide role-based access control to restrict unauthorized access to sensitive case data.
- Support audit logging, case tracking, and forensic report generation for investigation transparency.

1.5. Problem Statement

Traditional methods of storing evidence are often inefficient, vulnerable to tampering, difficult to organize, and challenging to maintain over long investigative periods. Physical storage systems can result in damaged evidence, poor accessibility, weak traceability, and limited accountability.

As investigations increasingly involve digital devices and electronic records, there is a growing need for a secure and reliable digital evidence management system. Existing manual approaches are not sufficient for handling large volumes of digital evidence while preserving authenticity and maintaining chain-of-custody records.

Indicium addresses these issues by providing a centralized platform that securely stores evidence, tracks all user activity, enforces authorization policies, and ensures integrity through cryptographic verification. This improves investigative efficiency while maintaining forensic reliability and accountability.

2. Overall Description

2.1. Product Perspective

Indicium is a standalone desktop application developed as a native Java-based forensic evidence management platform. It is designed specifically for secure digital evidence handling and investigative workflow management.

The system uses a layered architecture consisting of:

- Presentation Layer (JavaFX UI)
- Business Logic Layer
- Security & Authentication Layer
- Database Layer
- File Storage Layer

The system interacts with local file systems and databases to securely manage digital evidence while ensuring integrity and controlled access. Evidence files are stored in secure storage locations and linked to cases maintained in the database.

2.2. Product Functions

The major functions of Indicium include:

- User identity and access management
- Case creation and management
- Digital evidence ingestion and storage
- Cryptographic hashing and integrity verification
- Evidence viewing and secure downloading

- Cross-case evidence correlation
- Timeline reconstruction for investigations
- Bulk evidence import
- Audit logging and activity tracking
- Report generation
- System archiving and restoration
- Emergency lockdown management

3. Project Requirements

3.1. Software Requirement Specification

3.1.1. List of Features

- Secure evidence ingestion and storage with cryptographic hashing.
- Case creation, modification, and management for investigation workflows.
- Evidence categorization, filtering, and cross-case correlation mapping.
- Timeline reconstruction for chronological investigation tracking.
- Role-based access control for investigators, administrators, and auditors.
- Audit logging and chain-of-custody tracking for all system activities.
- Bulk evidence import and forensic report generation features.
- System archival, restoration, and emergency lockdown management.

3.1.2. Functional Requirements

3.1.2.1. Investigator Functions

- Create and manage investigation cases with relevant details and priority levels.
- Upload and review digital evidence files with metadata and integrity verification.
- Filter investigative data and link related evidence across multiple cases.
- Reconstruct investigation timelines and generate forensic reports.

3.1.2.2. Administrator Functions

- Manage user identities, permissions, and role assignments.
- Archive and restore investigation cases from long-term storage.
- Monitor audit logs and activate emergency lockdown procedures when required.
- Maintain system integrity and storage management operations.

3.1.2.3. System Functions

- Generate cryptographic hashes for uploaded evidence files.
- Verify evidence authenticity before viewing or downloading files.
- Record all user actions and evidence activities in audit logs.
- Prevent unauthorized access using role-based access control mechanisms.

3.1.3. Non-Functional Requirements

3.1.3.1. Performance Requirements

- Evidence uploads and integrity verification should complete with minimal delay.
- The system should efficiently handle bulk evidence imports and large file storage.
- Search and filtering operations should provide fast and accurate results.

3.1.3.2. Security Requirements

- All evidence files must remain read-only after upload to prevent tampering.
- Strong authentication and role-based authorization must be enforced.
- Audit logs and chain-of-custody records must remain protected and immutable.

3.1.3.3. Safety Requirements

- The system must preserve original evidence integrity.
- Read-only restrictions must prevent unauthorized modifications.
- Backup copies of modified records should be maintained.
- Emergency lockdown functionality should immediately stop all modification operations.

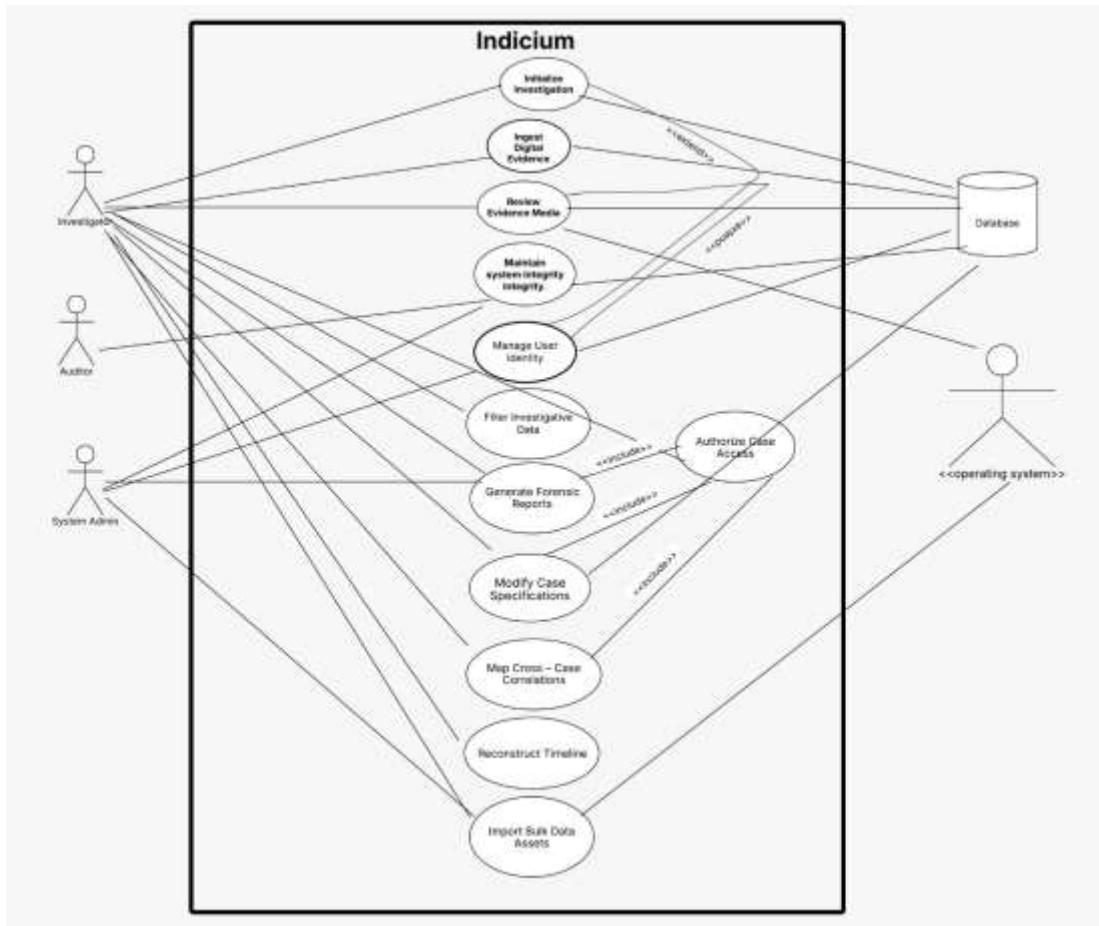
3.1.3.4. Reliability

- The system must prevent evidence corruption or loss during uploads and storage.
- Backup and archival mechanisms must ensure long-term evidence preservation.
- Interrupted uploads or transfer failures should not compromise stored evidence.

3.1.3.5. Scalability and Maintainability

- The system should support increasing evidence volumes and multiple active cases.
- Modular architecture should allow future integration of forensic tools and extensions.
- Database and storage management should remain efficient as data grows.

3.2. Use Case Diagram



3.3. High Level Use Cases

1. Manage User Identity
2. Filter Investigative Data
3. Initiate Investigation
4. Authorize Case Access
5. Ingest Digital Evidence
6. Modify Case Specifications
7. Review Evidence Media
8. Map Cross – Case Correlations
9. Reconstruct Timeline
10. Import Bulk Data Assets
11. Generate Forensic Reports
12. Manage System Integrity

3.4. Extended Use Cases

1. Manage User Identity

Use case Section	Comment	
Use case name	<i>Manage User Identity</i>	
Scope	Indicium – An Evidence Management System	
Level	Summary Level	
Primary actors	System Administrator	
Stake holders and interests	➤ Investigator: Needs a reliable account and appropriate permissions to access evidence and perform forensic tasks ➤ Auditor: Requires that every change to user identities (creation, modification, or deletion) is logged for security compliance. ➤ System Administrator: Wants an efficient way to manage system access while maintaining security protocols and database integrity.	
Main success scenario	User	Action
	1.	Admin selects “Identity Management” from the dashboard
	2.	Admin chooses to create a new account or modify an existing profile.
	4.	Admin enters user details (Name, Email and role (Investigator etc.))
	6.	Admin submits the changes to system
Extensions		System Response
	1.	System displays a list of current users and management options
	3.	System returns a form for filling relevant data
	5.	System validates that the email is unique and checks the roles are valid
	7.	System updates the Database
	8.	System records the transaction in the audit log.
Extensions	2a. Duplicate ID conflict:	
	➤ System alerts the admin that user already exists and stops account creation process.	
	3a. Security Policy Violation:	
	➤ System rejects the role assignment if they don’t follow the security rules.	
Extensions	4a. Database Connectivity Failure:	
	➤ System informs the admin of database failure	

	*a. Emergency Lockout: ➤ In the event of “all access closed” case, system rolls back changes and alerts the admin.
Pre conditions	➤ System Administrator is authenticated/authorized via RBAC rules. ➤ The indicium server is online and reachable.
Post Condition	➤ User identity records are created, updated, or deleted; all changes are committed to the database and logged in the system audit trail.

2. Filter Investigative Data

Use case Section	Comment	
Use case name	<i>Filter Investigative Data</i>	
Scope	Indicium – An Evidence Management System	
Level	User goal	
Primary actors	Investigator	
Stake holders and interests	Investigator – Wants to quickly narrow down relevant cases using specific filters System – Ensures search functionality is efficient and doesn’t expose unauthorized or irrelevant data	
Main success scenario	User	Action
	1. Investigator opens the case search panel	2. System displays the filter panel with available filter options
	3. Investigator selects the required filter	4. System validates the filter
	5. Investigator applies the filter	6. System executes the filter and retrieves the matching cases
	7. Investigator reviews the filtered results	8. System displays the filtered cases in a presentable way
Extensions	3a. No filter is selected: ➤ System displays all the cases as it is 8a. No cases displayed: ➤ System prompts that there are no cases matching the applied filter(s)	
Pre-Conditions	➤ Investigator is authenticated and has an active session. ➤ There is at least one record in system to apply filter(s) on.	
Post conditions	➤ Cases matching the filtered criteria are displayed. ➤ No unauthorized or irrelevant records and exposed.	

3. Initiate Investigation

Use case Section	Comment	
Use case name	<i>Initiate Investigation</i>	
Scope	Indicium – An Evidence Management System	
Level	User goal	
Primary actors	Lead Investigator	
Stake holders and interests		
Main success scenario	User Action	System Response
	9. Investigator selects the option to start a new investigation.	2. System verifies that the user has a create case permission.
		3. System displays a blank case initialization form.
	4. Investigator enters the core details (Case Title, Incident Date, Description, and Priority Level).	
	5. Investigator submits the case.	6. System validates that all required information is present and correct.
		7. System generates a unique, non-sequential Case ID.
		8. System displays a success message with a unique case Id.
Extensions	1a. User lacks required permissions (RBAC Failure)	
	➤ System detects the user does not have the case creator role (e.g., user is a Read-Only Analyst). ➤ System denies access to the form. ➤ System displays an error message. ➤ System logs the unauthorized attempt for security review.	
	4a. Investigator cancels the process	
	➤ Investigator clicks "Cancel" at any point before submission. ➤ System discards any entered text and returns to the dashboard	
	6a. Missing required information:	
	➤ System highlights the missing fields (e.g., no Incident Date). ➤ System prompts the Investigator to complete them.	
	6b. Case Title already exists:	
	➤ System detects a duplicate case name.	

	➤ System asks Investigator to confirm if this is a duplicate or to provide a unique variation.
Pre conditions	➤ The Investigator is logged into the system. ➤ The Investigator has case creation permissions.
Post conditions	➤ A new case is created in the system with a unique ID. ➤ The system is ready to accept evidence for this case. ➤ The creation event is logged in the system history.

4. Authorize Case Access

Use case Section	Comment	
Use case name	Authorize Case Access.	
Scope	Indicium – An Evidence Management System	
Level	Subfunction Level	
Primary actors	Investigator	
Stake holders and interests	➤ Investigator: Needs timely access to case data to proceed with an investigation. ➤ System Administrator: Needs to ensure only authorized personnel are accessing sensitive forensic data. ➤ Legal Teams: Require that access is strictly controlled to maintain the chain of custody and legal admissibility. ➤ Auditor: Needs a timestamped record of who accessed which case and when.	
Main success scenario	User Action	System Response
	1. Calling use case requests authorization for a specific case ID on behalf of the investigator	2. System identifies the target Case ID and Investigator ID.
		3. System verifies the investigator has standard viewing privileges.
		4. System grants the Investigator Read Only Access.
		5. System writes the timestamped entry into logs file and notifies the admin
Extensions	2a. Insufficient Privileges:	

	➤ If investigator doesn't have sufficient view privileges, the system halts the process. 4a. Audit Log Failure: ➤ In case of log file failure, system terminates the authorization request. *a. Security Lockdown: ➤ In case that "Close All Access" flag is active, system denies authorization.
Pre conditions	➤ The user is authenticated and active in the system. ➤ The target case ID exists in the system.
Success Guarantee	➤ The user is granted access to the case and its contents for a secure timed session. ➤ The outcome of the session is also recorded and admin is notified.

5. Ingest Digital Evidence

Use case Section	Comment	
Use case name	<i>Ingest Digital Evidence</i>	
Scope	Indicium – An Evidence Management System	
Level	User-Goal	
Primary actors	Investigator	
Stake holders and interests	Defence Attorney	
Main success scenario	User Action	System Response
	1. Investigator opens the specific case file within the system.	
	2. Investigator selects the add evidence option.	
		3. System presents all evidence addition options (images, videos, document, bit stream ingestion etc.)
	4. Investigator selects the required option.	5. System presents the option to add files to the queue.
	6. Investigator selects the files from their local drive	

	or drags them into the upload area.	
	7. Investigator adds context details (Source of evidence, Description, Date Seized) and confirms.	
		8. System uploads the data while automatically generating a unique digital fingerprint.
		9. System saves the evidence file and locks it so it cannot be edited.
		10. System updates the case log and shows verified status.
Extensions	8a. Transfer interrupted (e.g., USB pulled out): ➤ System detects an incomplete transfer. ➤ System deletes the partial file fragment to keep storage clean. ➤ System alerts the Analyst to retry. 8b. Duplicate Evidence: ➤ System notices the digital fingerprint matches a file already in the case. ➤ System prompts the Analyst to confirm if this is a duplicate copy or a mistake.	
Pre conditions	➤ A valid Case ID exists. ➤ The Analyst has the digital files ready on a local drive.	
Post conditions	➤ The evidence is securely stored in the system. ➤ A unique digital fingerprint is generated to prove authenticity. ➤ The activity log is updated to show who uploaded the evidence.	
Special Requirements	➤ Performance: The Integrity Check (hashing) must happen in under 2 seconds for average-sized files, so the user doesn't get frustrated.	
Technology & Techniques	➤ Hashing Algorithms (SHA-256): The system must have a built-in calculator that scans the file bit-by-bit to generate a unique digital fingerprint. This <i>must</i> happen locally or immediately upon receipt. ➤ Virus Scanning API: An integrated security tool that scans incoming files for malware before they are saved to the main server.	

6. Modify Case Specifications

Use case Section	Comment	
Use case name	Modify Case Specifications	
Scope	Indicium – An Evidence Management System	
Level	User goal	
Primary actors	Case Manager, Investigator	
Stake holders and interests	Investigator – Needs to update case details Admin – Ensure data integrity System – Must log all modifications with timestamps and user IDs for chain of custody and save copies of previous data	
Main success scenario	User Action	System Response
	1. Investigator selects the existing case from Cases module	2. System loads details of the selected case
	3. Investigator selects the Edit option and waits for system to authenticate.	4. System verifies the user, If authorized, editing becomes permissible
	5. Investigator modifies the required fields in the case	6. System highlights the modified fields.
	7. Investigator saves and submits the changes	8. System stores a copy of previous data and updates the case with modifications made.
Extensions	4a. Investigator is not authorized to modify the case: ➤ System prompts “Access Denied” and keep the details read – only	
	5a. Investigator attempts to modify locked fields: ➤ System prevents editing of those fields ➤ Notifies the user that they are locked by the admin	
	7a. Empty field error: ➤ System highlights the field that is left empty ➤ Prompts the user to fill it	
	8a. Database write error: ➤ System rolls back the attempted update, logs the failure and notifies the user	
Pre conditions	➤ Investigator is authenticated and has an active session. ➤ Investigator hold read/write permission. ➤ Targeted case exists and is not locked or archived	
Post conditions	➤ Modifications in the case are updated. ➤ An audit log entry is created recording the change. ➤ Copy of the case without the modifications done is saved as backup. ➤ All users with access to the case see the updated information on next load.	

7. Review Evidence Media

Use case Section	Comment	
Use case name	<i>Review Evidence Media</i>	
Scope	Indicium – An Evidence Management System	
Level	User goal	
Primary actors	Investigator	
Stake holders and interests	Internal Affairs, IT Security	
Main success scenario	User	Action
	1. Investigator browses the evidence list for a specific case.	
	2. Investigator selects a specific file (e.g., a photo or document).	3. System performs a background check to ensure the file's current digital fingerprint matches the original upload hash.
		4. System confirms the file is authentic.
		5. System opens the file in the secure Player/Viewer (read-only mode).
	6. Investigator examines the evidence within the application.	
		7. System records the view entry in the audit log.
Extensions	2a. Investigator chooses to Download: ➤ Investigator selects the file and chooses the download option. ➤ System shows a warning that this action is tracked for legal reasons. ➤ Investigator confirms the action. ➤ System checks the digital fingerprint of the file to ensure the copy is exact. ➤ System saves the file to the Investigator's computer. ➤ System records the download in the activity log to distinguish it from a simple view.	
	3a. Verification Check Fails (Tampering Detected): ➤ System notices the file's digital fingerprint does not match the original record. ➤ System stops the file from opening or downloading. ➤ System shows a warning that the evidence might be changed or damaged. ➤ System marks the file for the Administrator to check.	

	5a. File type cannot be viewed in the application: ➤ System cannot show this specific type of file (like specialized security camera video) in the application viewer. ➤ System asks the Investigator to download the file instead to view it on their computer. 2a, Step 5a. Download Interrupted (Connection Lost): ➤ System sees the internet connection broke during the download. ➤ System stops the transfer. ➤ System tells the user the download failed and asks them to try again. (The system does not record this failed attempt in the official log).
Pre conditions	➤ The investigator has read access. ➤ The evidence to be viewed has been successfully uploaded.
Post conditions	➤ The user has viewed or downloaded the evidence. ➤ The system has verified the evidence is still original. ➤ An access log entry is created specifying the action type (View or Download).
Technology & Techniques	➤ Native Operating System Integration Indiciu utilizes Java's native interface capabilities to interact directly with the host operating system to execute the appropriate system commands. This allows Indiciu to launch the default associated application for reviewing specific evidence types (such as video players or pdf viewers) without requiring embedded third-party codecs.

8. Map Cross – Case Correlations

Use case Section	Comment	
Use case name	Map Cross-Case Correlations	
Scope	Indiciu – An Evidence Management System	
Level	User Goal Level	
Primary actors	Investigator	
Stake holders and interests	➤ Investigator: Wants to identify patterns and links between different pieces of evidence across multiple cases without modifying the original files. ➤ System Administrator: Ensures that cross-case linking follows authorization protocols. ➤ Auditor: Needs to track the creation of links between cases for investigative transparency.	
Main success scenario	User Action	System Response
	1. Investigator selects a specific piece of evidence in their current workspace	2. System prompts the investigator to search for target case.

	and chooses the option of creating a cross-case link.	
	3. Investigator inputs search data and selects the target case	4. System triggers a “Authorize Case Access” event for target case.
	5. Investigator browses target case and selects the evidence to link with.	6. System links the two evidence objects together.
	7.	8. System logs a timestamped entry to audit log and updates the objects.
Extensions	3a. Authorization Failure for Target Case: ➤ If the subfunction determines the investigator ineligible for access to case B: 1. The system halts the mapping process 2. Notifies the admin for approval. 5a. Existing Link Conflict: ➤ If the link already exists 1. It rejects the submission. 2. Notifies the investigator.	
Pre conditions	➤ Investigator is authenticated. ➤ Investigator is currently viewing an authorized piece of evidence	
Success Guarantee	➤ A secure connection record is established between two or more separate evidences, the link is logged and the original file remains unmodified.	

9. Reconstruct Timeline

Use case Section	Comment
Use case name	<i>Reconstruct Incident Timeline</i>
Scope	Indicium – An Evidence Management System
Level	User goal
Primary actors	Investigator
Stake holders and interestsI	Investigator – Needs to build and refine a chronological sequence of events to support case analysis and reporting. Admin – Wants to review and approve timeline before submission. Auditor – Requires an accurate, tamper-evident account of the incident chronology. System – Must associate timeline entries with existing evidence records to preserve traceability.

	User Action	System Response
Main success scenario	1. Investigator selects the case and navigates to timeline section/tool.	2. System loads any existing timeline entries for the case in chronological order.
	3. Investigator adds a new event.	4. System presents a form with fields to add and option to link evidences.
	5. Investigator fills in the event details and optionally links one or more evidence items.	6. System validates the date/time format and checks that linked evidence belongs to the same case.
	7. Investigator saves the event.	8. System updates the timeline chronology with the event added and saves the timestamp with user ID.
	9. User repeats steps 2–4 to add additional events as needed.	10. System continuously updates the chronological view and maintains internal sort order by timestamp.
	11. Investigator clicks an existing event to edit it.	12. System opens the event in editable form.
	13. Investigator modifies the event and saves it.	14. System updates the record, logs the change in the audit trail, and refreshes the timeline.
Extensions	5a. Invalid date/time format entered: ➤ System highlights the field and prompts for a correct format; event is not saved. 5b. Linked evidence does not belong to the current case: ➤ System rejects the association and notifies the user. 8a. Two events share the same timestamp: ➤ System saves both and orders them by insertion sequence within the same timestamp. 11a. User attempts to delete a timeline event: ➤ System prompts for confirmation: ➤ If yes, then removes the entry and logs the deletion in the audit trail. ➤ If no, cancels deletion and return to timeline view	
Pre conditions	➤ User is authenticated with Write access to the case. ➤ The target case exists and is in an active state. ➤ At least one evidence item is ingested into the case (to support optional evidence linking).	
Post conditions	➤ One or more chronological timeline events are saved and associated with the case. ➤ Timeline entries that link evidence maintain a traceable reference to those evidence records.	

- All additions and edits are recorded in the system audit log.

10. Import Bulk Data Assets

Use case Section	Comment	
Use case name	Import Bulk Data Assets	
Scope	Indicium – An Evidence Management System	
Level	System sub function	
Primary actors	Investigator, Admin	
Stake holders and interests	Investigator – Needs to ingest large volumes of evidence files from external drives or directory structures quickly and reliably. Admin – Wants to confirm that all imported assets are correctly catalogued and linked to the right case. System – Must compute and store cryptographic hashes for every imported file to guarantee integrity. Auditor – Requires an import log showing which files were ingested, by whom, and when.	
Main success scenario	User Action	System Response
	1. Investigator opens a case and selects Import Bulk Data from the Evidence menu.	2. System displays the Bulk Import dialog with options to select a source file or directory.
	3. Investigator selects source folder/files.	4. System reads the selected files and presents a preview with their name, size and type.
	5. Investigator reviews the preview and selects Import	6. System begins the import process.
		7. System import processes each file, storing the binary content, metadata, hash value, file type, and originating source path.
		8. System completes import, generates an import summary report and records the event in the audit log.
	9. Investigator views the import summary	10. System displays the summary and returns the user to the case Evidence view with newly imported items listed.

Extensions	3a. Selected path is empty or inaccessible: ➤ System displays an error message and returns the user to source selection. 5a. Selected file set exceeds storage quota: ➤ System aborts the import and notifies the user of the limit. 7a. A file fails hash computation or read error occurs: ➤ System skips the file, logs the failure, and continues processing remaining files. ➤ Failed items are flagged in the import summary. 7b. Duplicate detected (same hash already exists in the case): ➤ System notifies the user and prompts the user to keep/skip the duplicate based on preference.
Pre conditions	➤ User is authenticated with Write access to the target case. ➤ The target case exists and is in an active state. ➤ The source files or directory are accessible to the system. ➤ Sufficient storage is available for the import.
Post conditions	➤ All successfully imported files are stored as evidence records in the database with verified cryptographic hashes. ➤ Each evidence record is linked to the current case. ➤ An audit log entry captures the import event including user, timestamp, and item count. ➤ The import summary is available for review.

11. Generate Forensic Reports

Use case Section	Comment	
Use case name	<i>Generate Forensic Reports</i>	
Scope	Indicium – An Evidence Management System	
Level	User Goal Level	
Primary actors	Investigator/System Admin	
Stake holders and interests	Legal System: Requires highly accurate, formatted, and read-only reports that are admissible in court System Admin: Needs to ensure that report generation does not compromise system performance or security. Investigator: Needs to quickly extract relevant data into a professional format for presentation	
Main success scenario	User Action	System Response
	1. Actor selects the case and report type (“case summary” or “System Audit log”).	2. System prompts the actor to define the parameters via a form.

	3. Actor enters the required data.	4. System triggers “Authorize Case Access” subfunction.
	5.	6. System gathers all the relevant data from the database and sends the data for conversion to relevant file format.
	7. Actor selects a save option.	8. System passes the formatted report to the user directory and updates logfile.
Extensions	2a. Permission Denied: ➤ In case of insufficient privileges, system halts the process. 3a. Empty Dataset: ➤ If empty dataset: ran 1. System halts the process 2. Notifies actor.	
Pre conditions	➤ Actor is authenticated and active ➤ The target data exists in the system.	
Success Guarantee	➤ A read only, formatted document is successfully exported to the system to be viewed or downloaded.	

12. Manage System Integrity

Use case Section	Comment	
Use case name	<i>Manage System Integrity</i>	
Scope	Indicium – An Evidence Management System	
Level	System Admin Level	
Primary actors	System Administrator	
Stake holders and interests	Department of crime and justice (Police)	
Main success scenario	User Action	System Response
	1. Administrator opens the system maintenance dashboard.	
		2. System displays options to Archive, Restore, or Lock Down the system

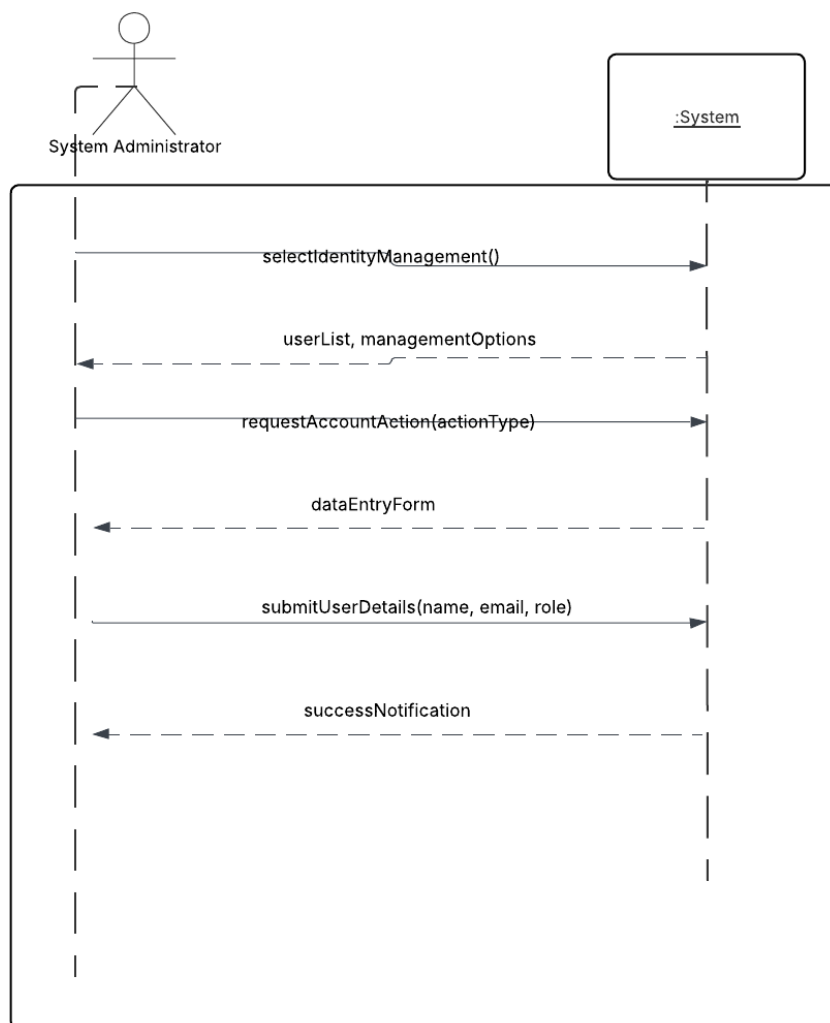
	3. Administrator selects option to archive an old case.	4. System shows a list of cases that have been closed for a long time.
	5. Administrator selects the cases to move and confirms the action.	6. System checks the digital footprint of files to see that they are right or not.
		7. System moves the files to long term storage.
		8. System removes the files from active space to free up storage.
		9. System records the action in the activity log.
Extensions	3a. Administrator restores an old case: ➤ Administrator selects the option to restore data. ➤ System asks for the case number. ➤ Administrator enters the number and confirms. ➤ System retrieves the files from long-term storage. ➤ System compares the current digital fingerprint with the original one to ensure nothing changed while it was stored. ➤ System adds the case back to the active list for investigators to see. 3b. Administrator activates the Emergency Lockdown (Kill Switch): ➤ Administrator selects the emergency lockdown button. ➤ System asks for a second password or security code to confirm this serious action. ➤ Administrator enters the code. ➤ System immediately logs out all other users. ➤ System stops anyone from adding or changing data. ➤ System displays a warning message on the login screen saying the system is unavailable. ➤ System sends an alert to the security team. 6a. Archiving fails due to full storage: ➤ System sees that the long-term storage area is full. ➤ System stops the transfer. ➤ System alerts the Administrator that there is no space left. ➤ System ensures the original files are kept safe in the main list.	
Pre conditions	➤ The Administrator is logged into the system. ➤ The Administrator has special permission to manage system safety.	
Post conditions	➤ Archiving: Old cases are moved to long-term storage to save space. ➤ Restoring: Old cases are brought back to the main screen for review. ➤ Kill Switch: The system is locked down, and all users are logged out.	
Special Requirements	➤ Reliability: The system must never lose a file once the upload is confirmed. ➤ Security: All evidence files must be "Read-Only" for standard users; they can never be modified, only viewed.	

3. System Overview

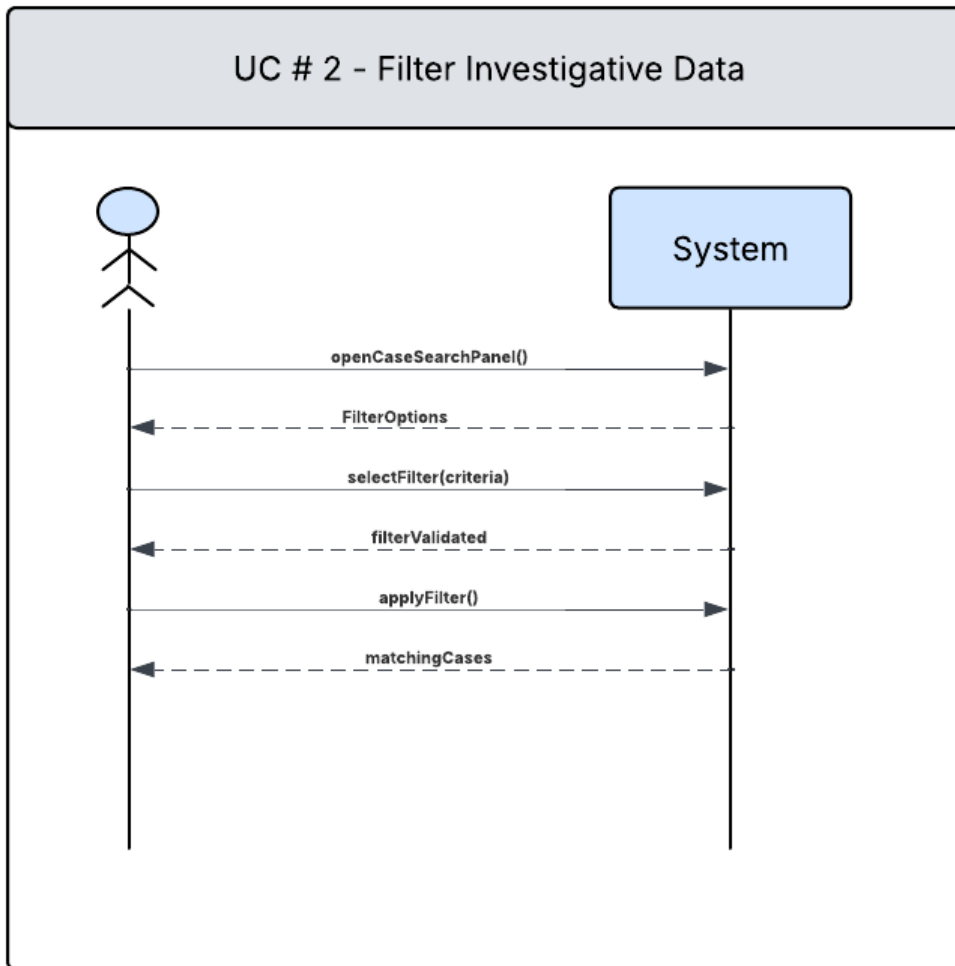
3.1. System Sequence Diagram

1. Manage User Identity

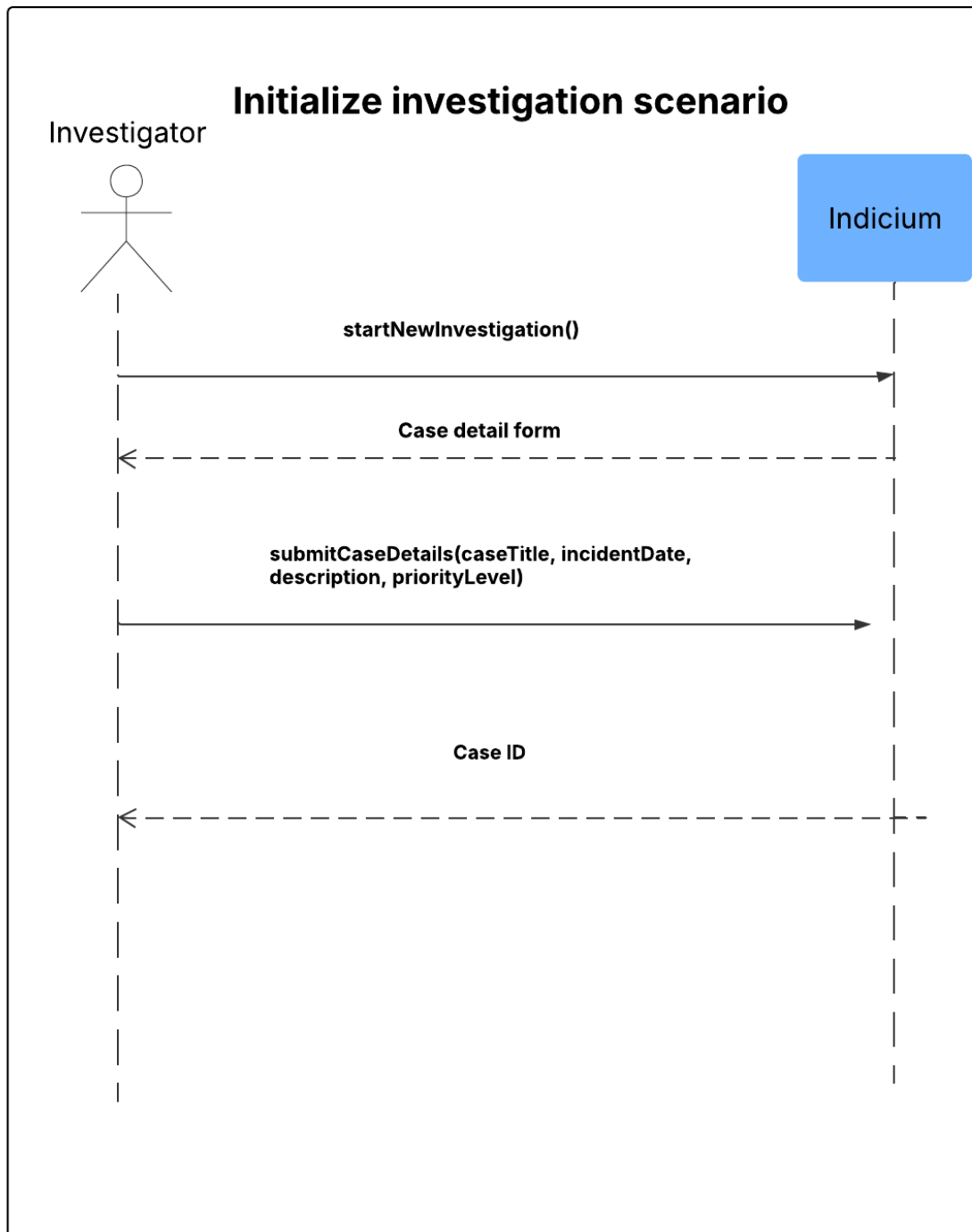
Manage User Identity



2. Filter Investigative Data

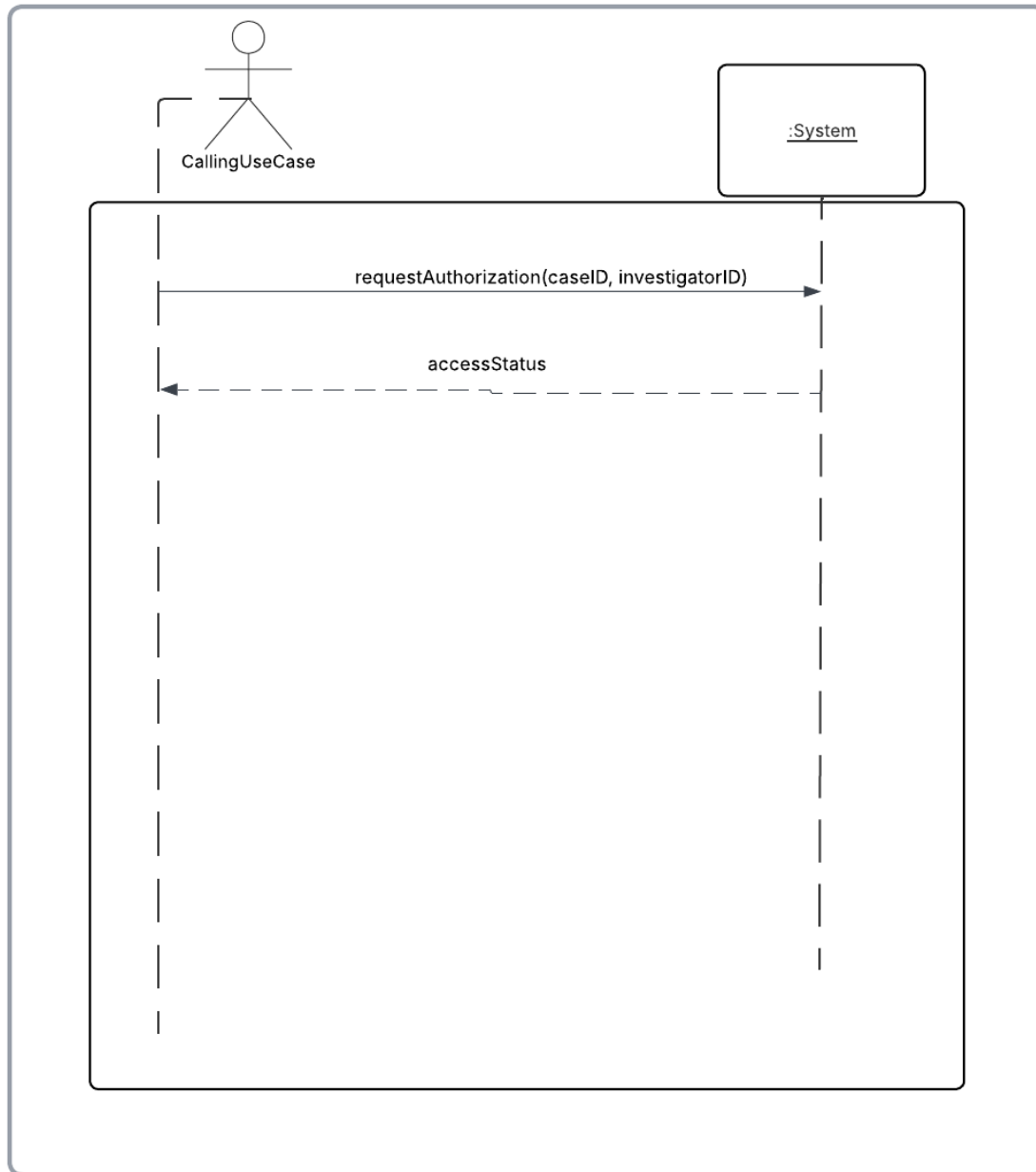


3. Initiate Investigation

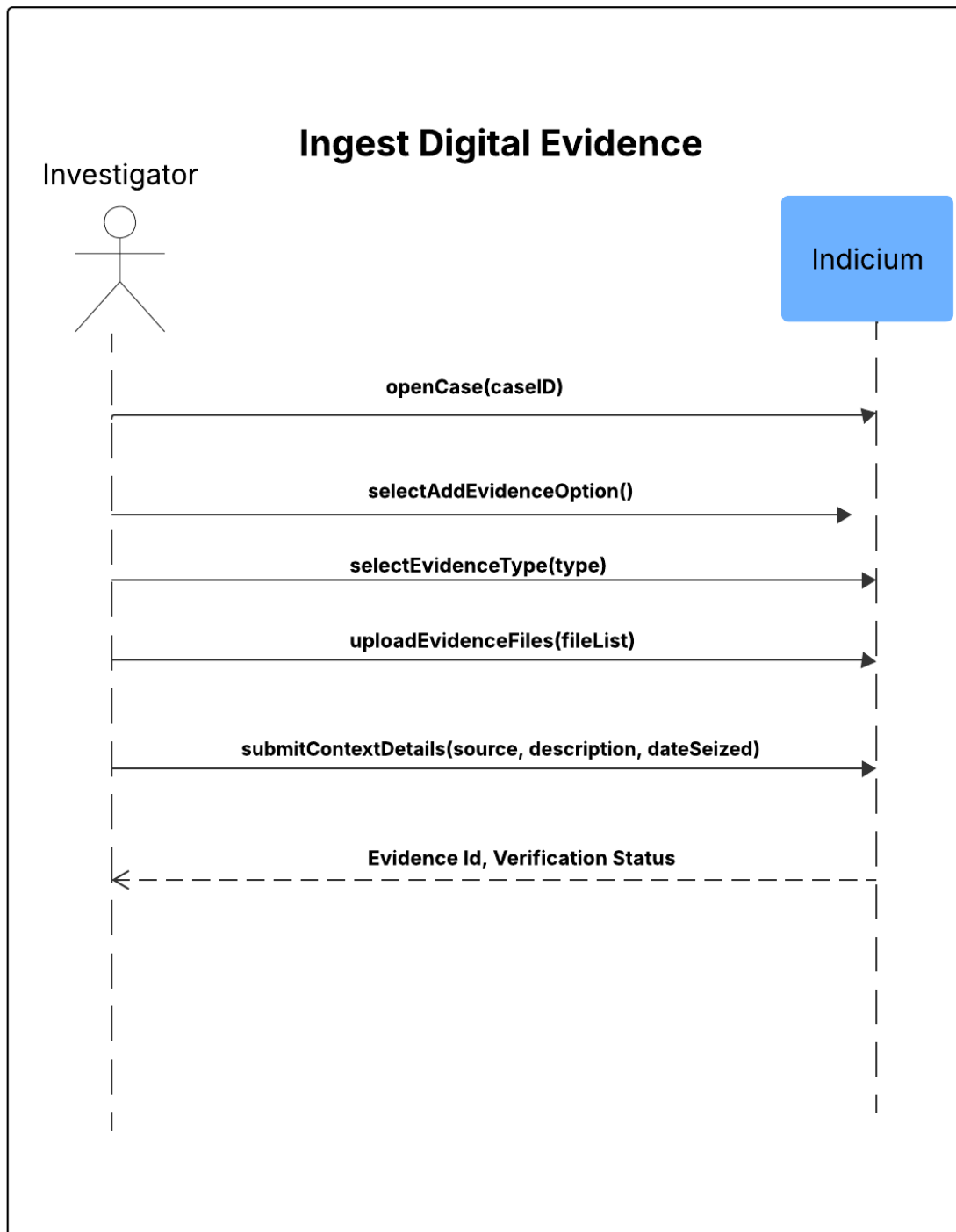


4. Authorize Case Access

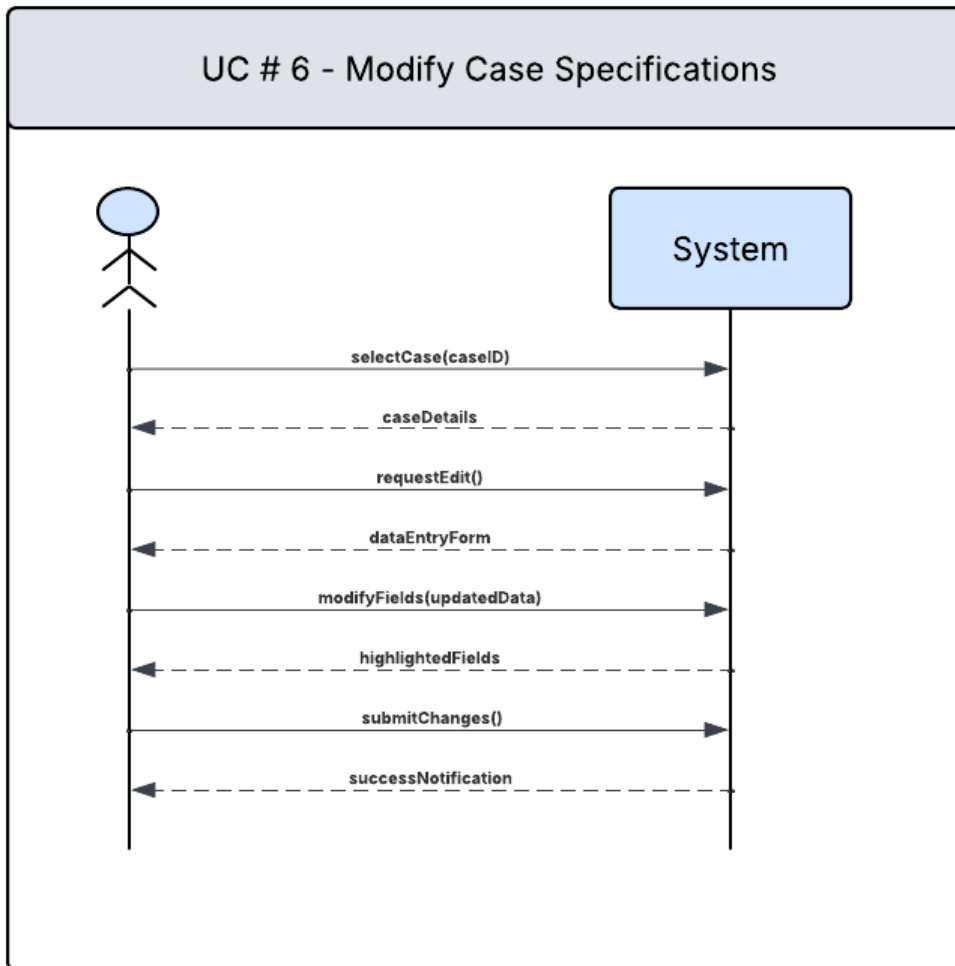
Authorize Case Access



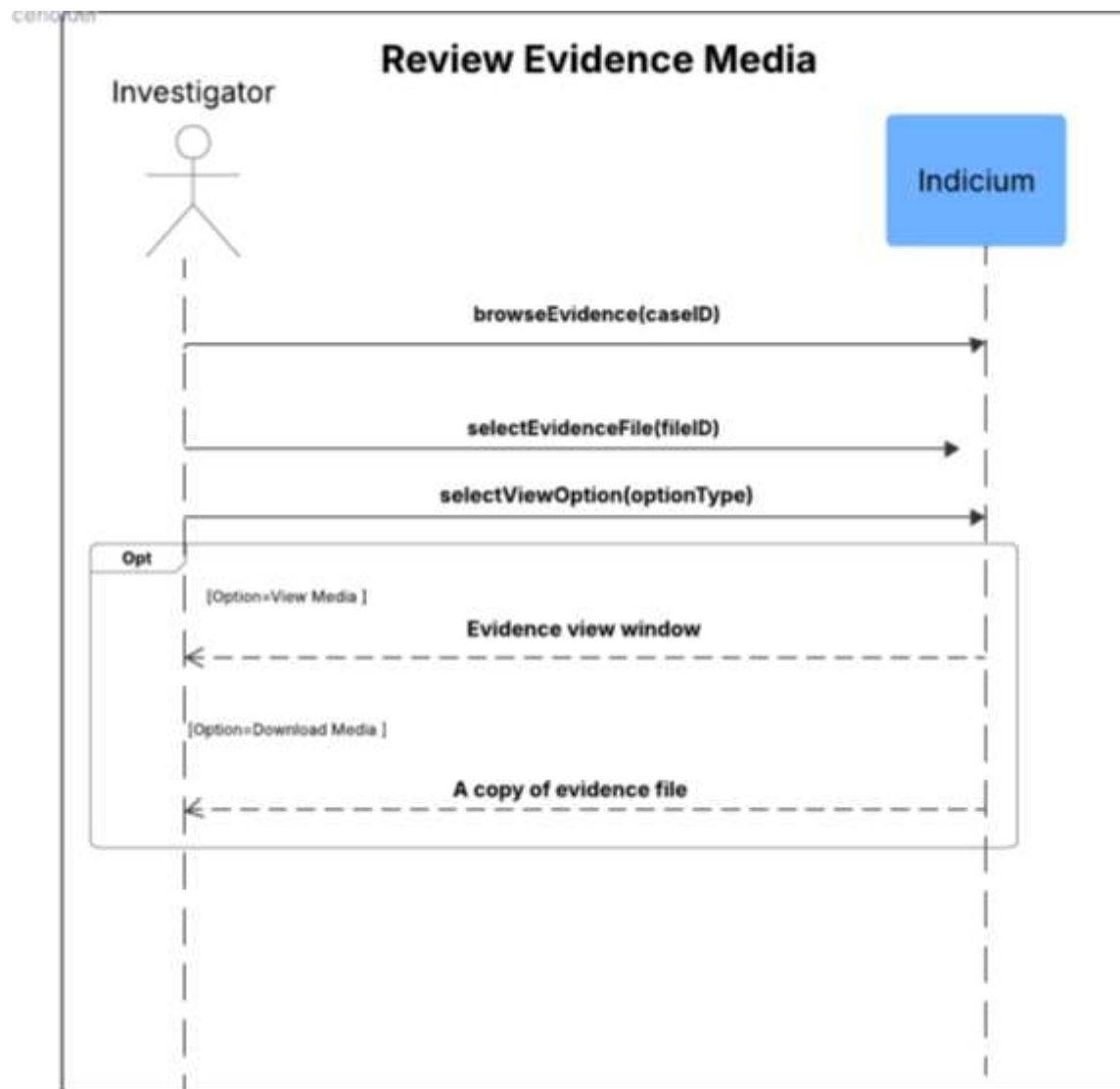
5. Ingest Digital Evidence



6. Modify Case Specifications

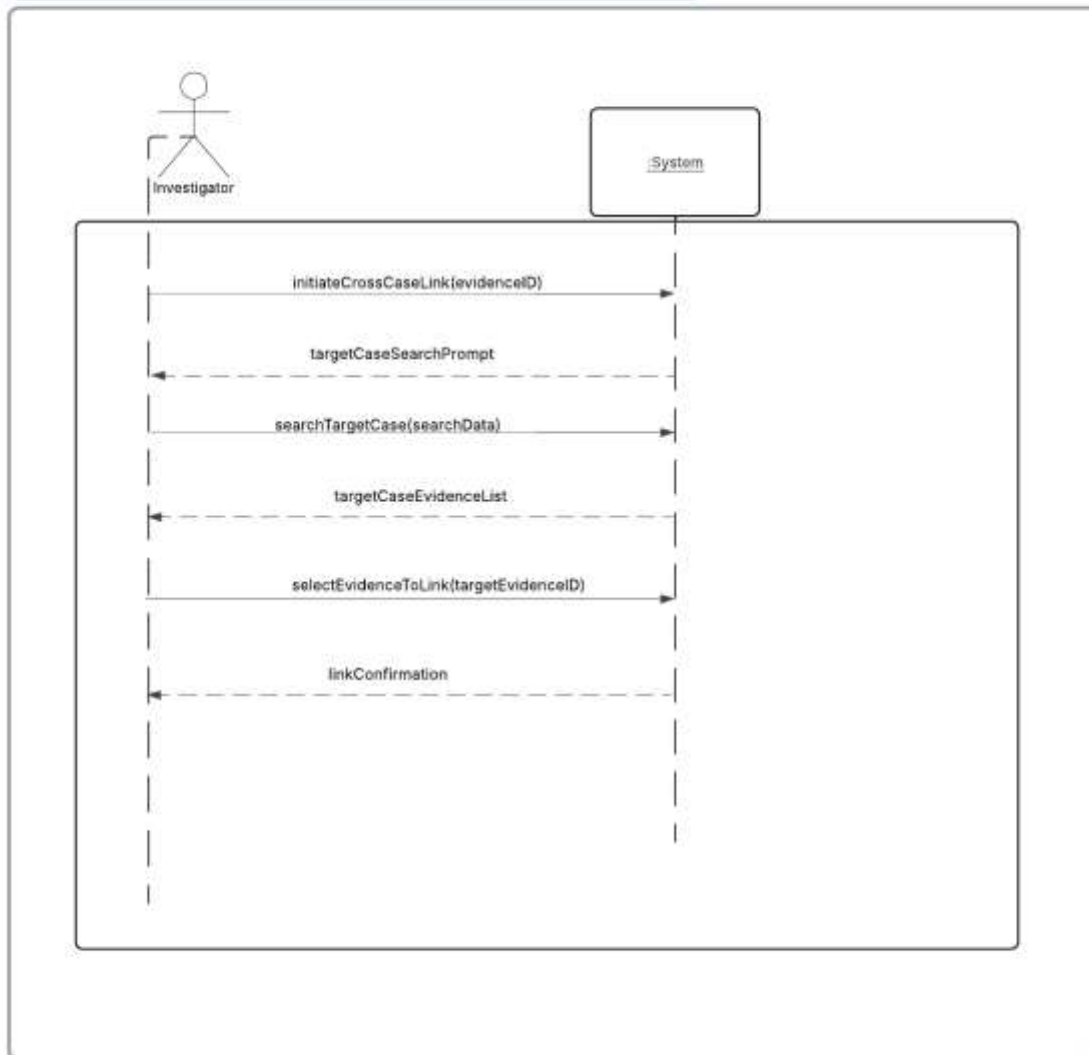


7. Review Evidence Media

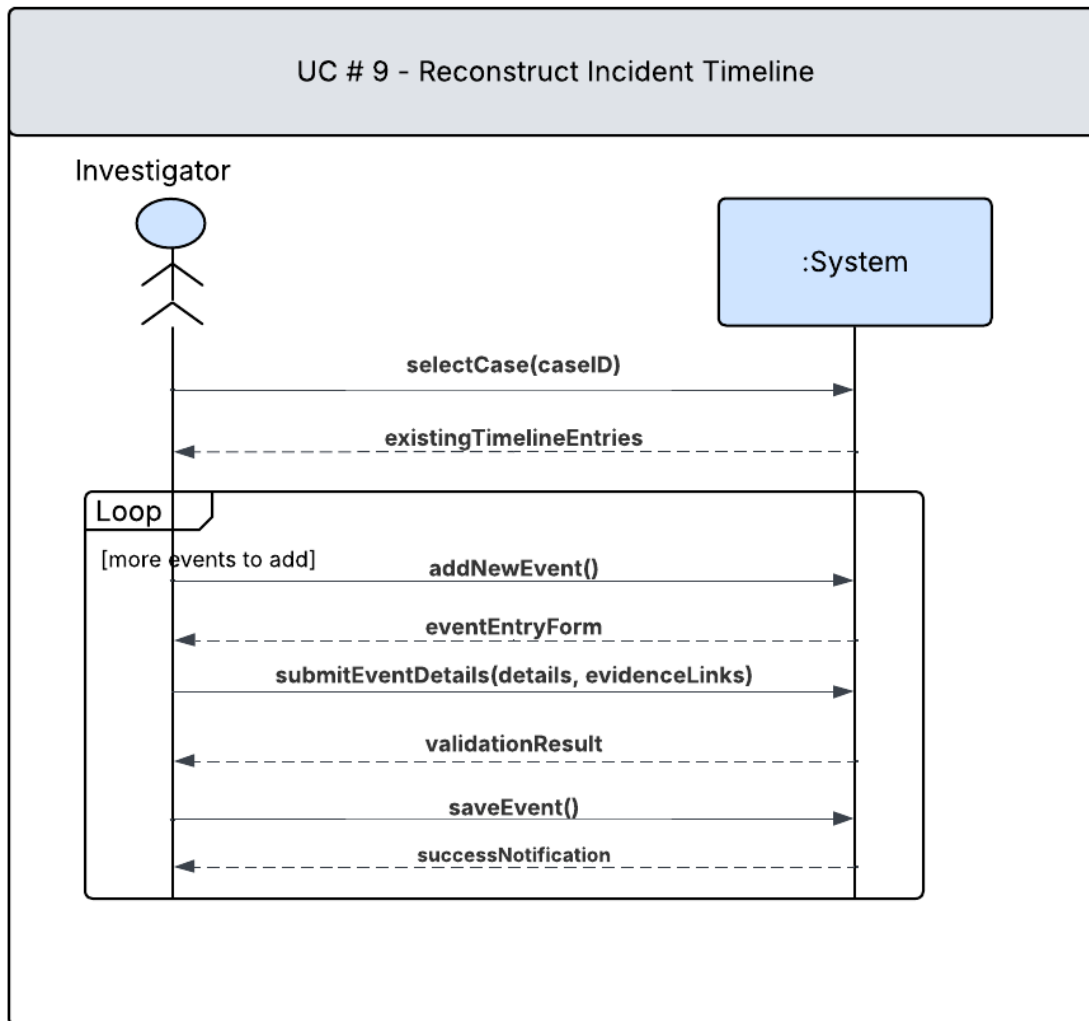


8. Map Cross – Case Correlations

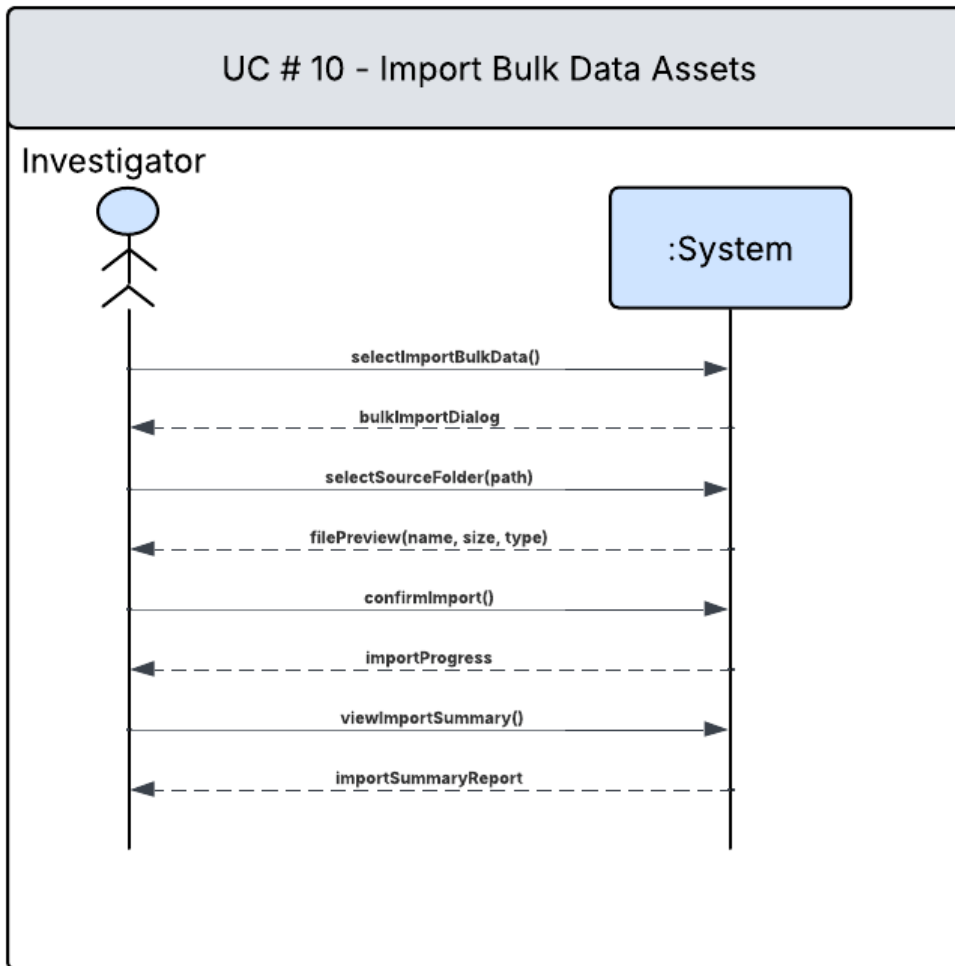
Map Cross-Case Correlations



9. Reconstruct Timeline

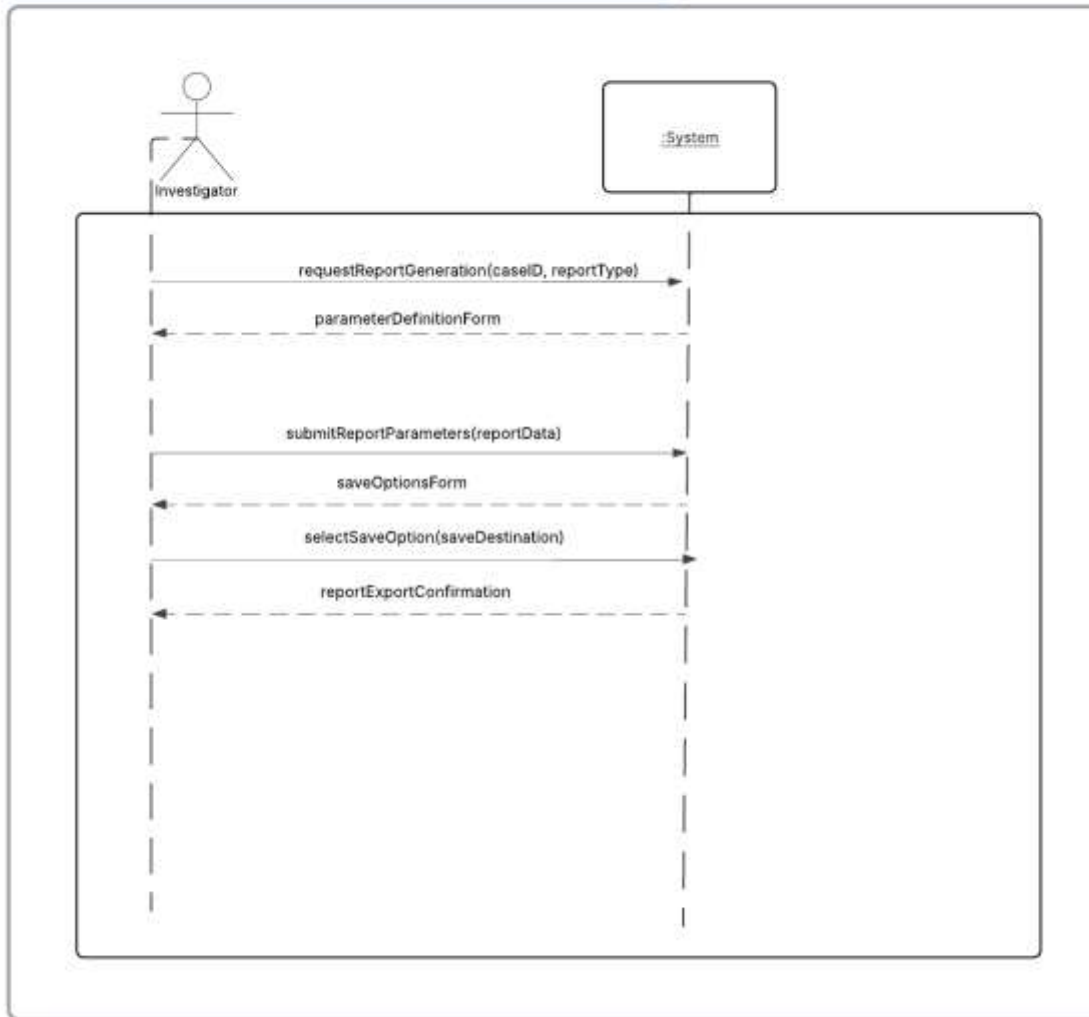


10.Import Bulk Data Assets

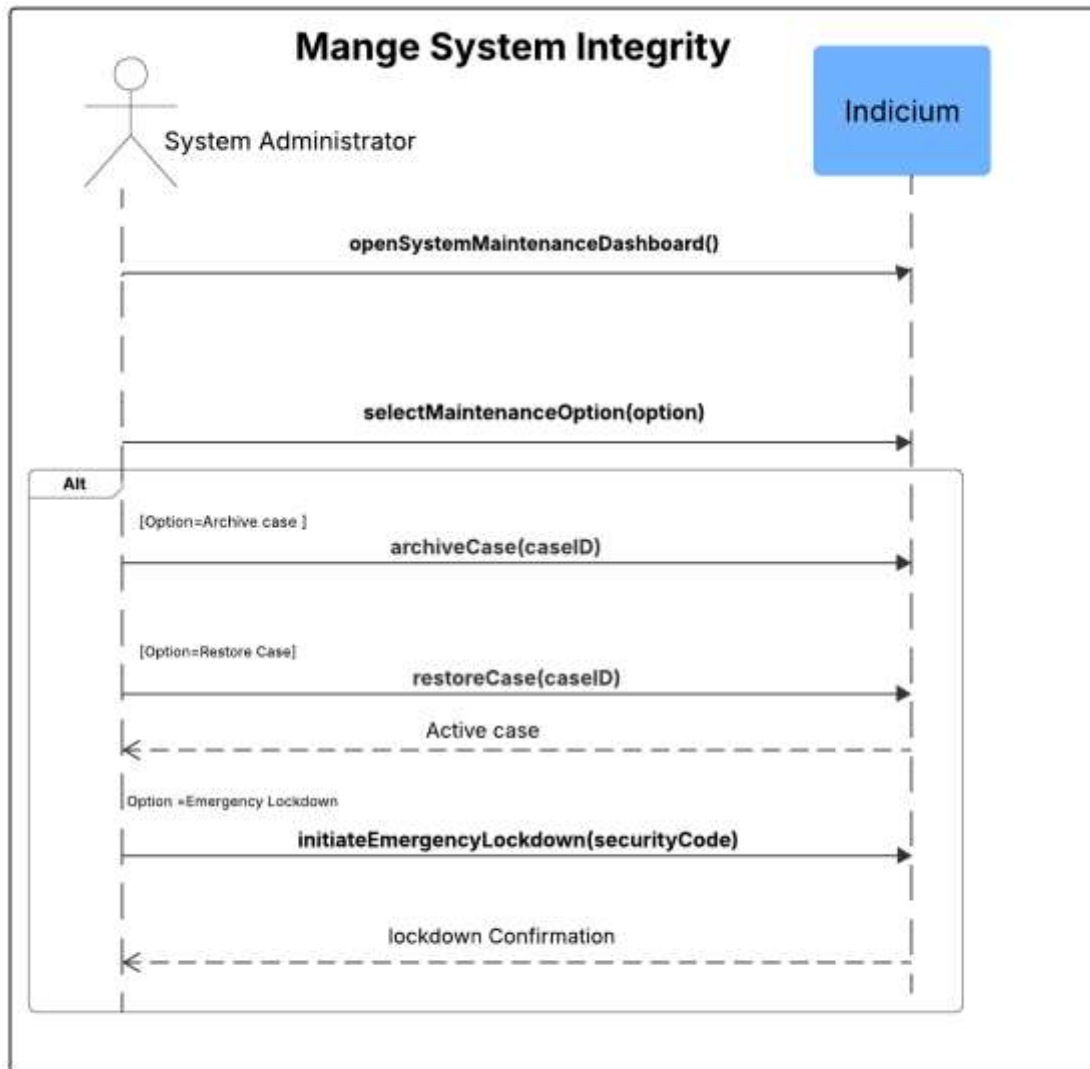


11. Generate Forensic Reports

Generate Forensic Reports

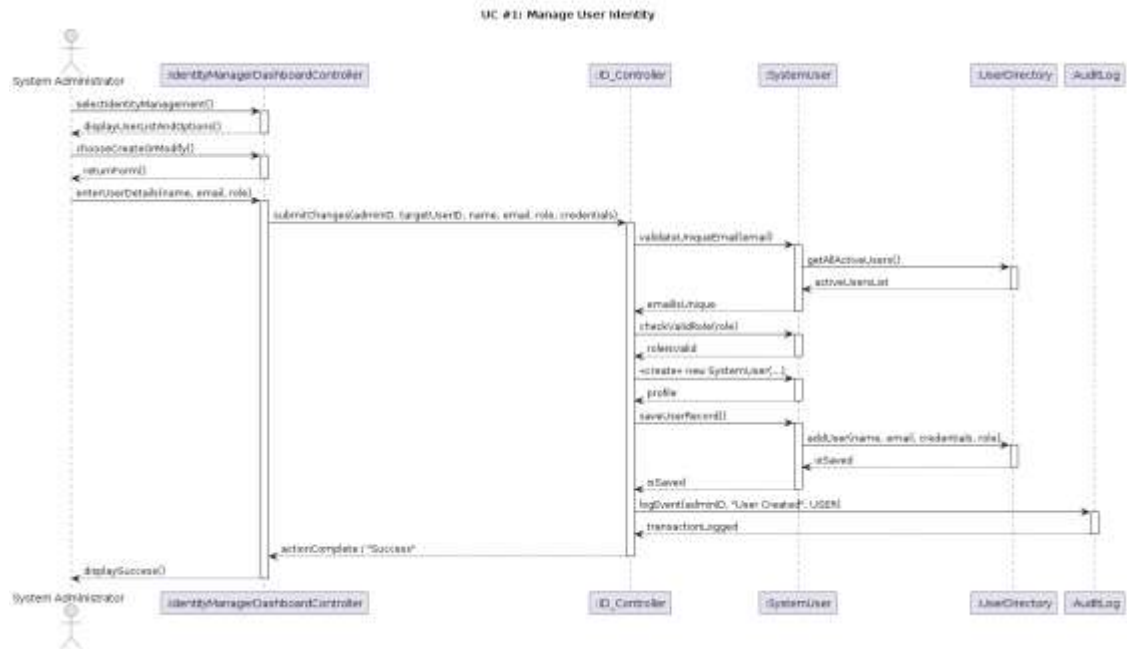


12. Manage System Integrity

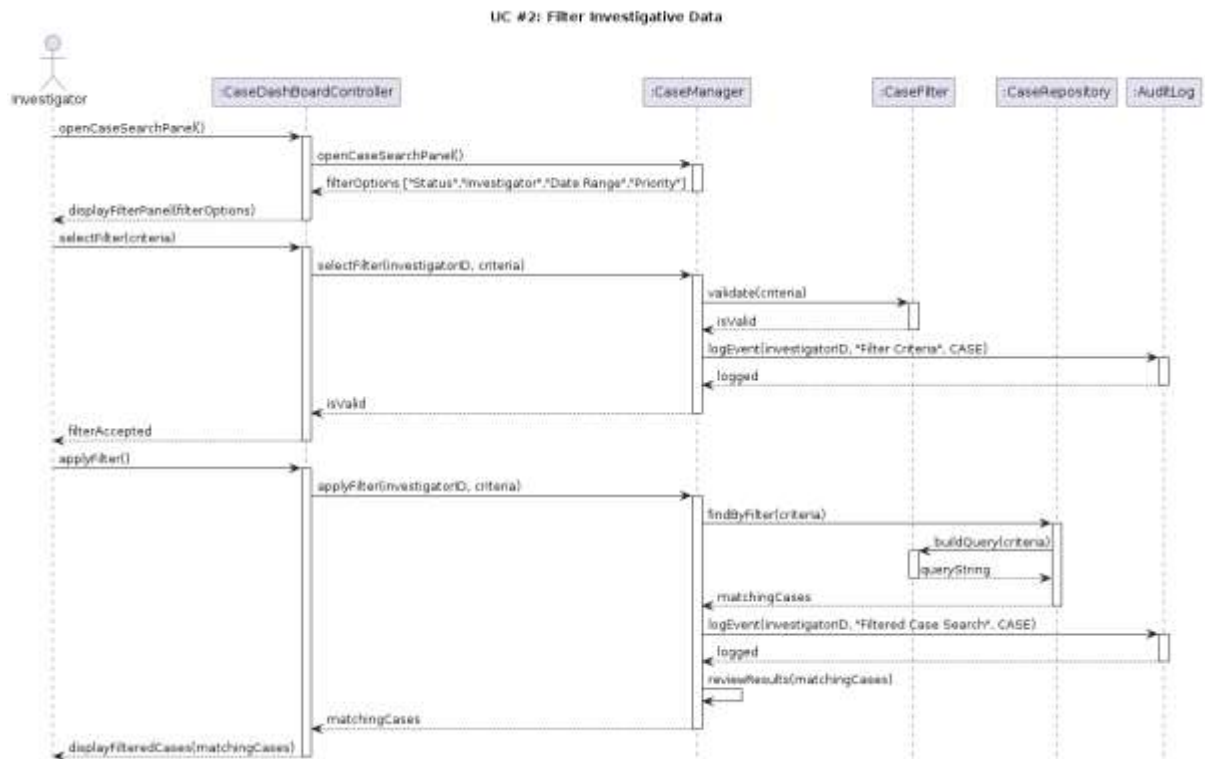


3.2. System Diagrams

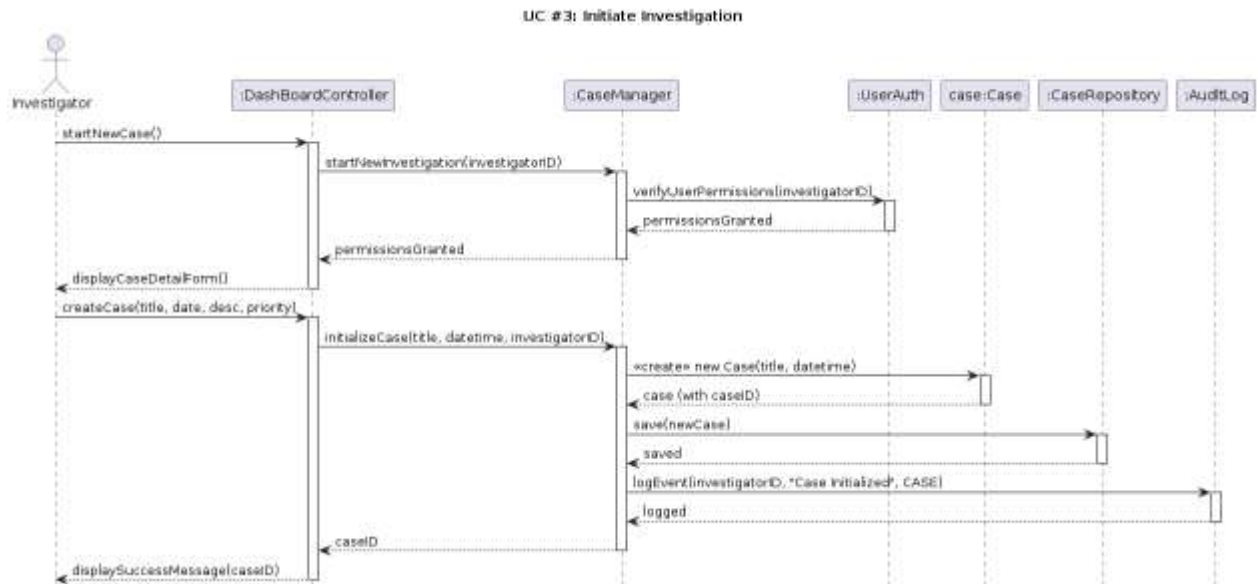
1. Manage User Identity



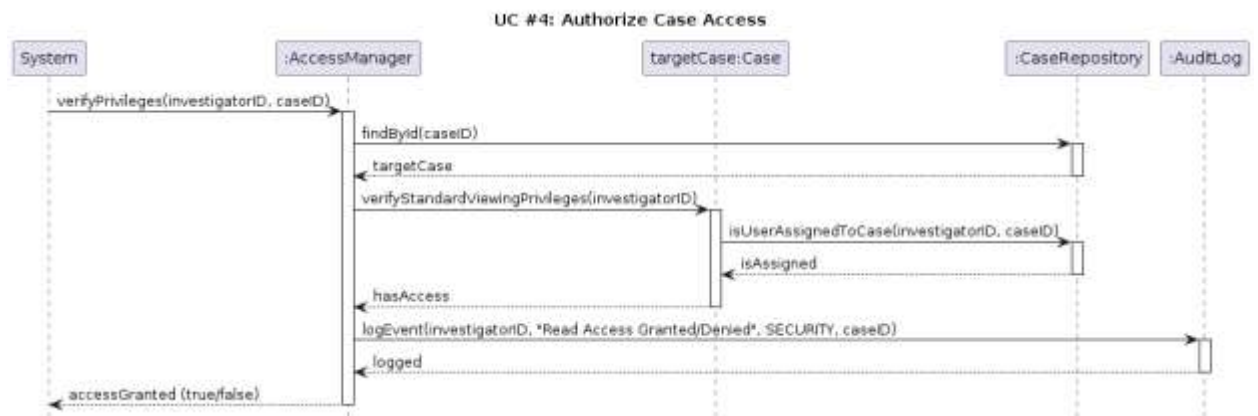
2. Filter Investigative Data



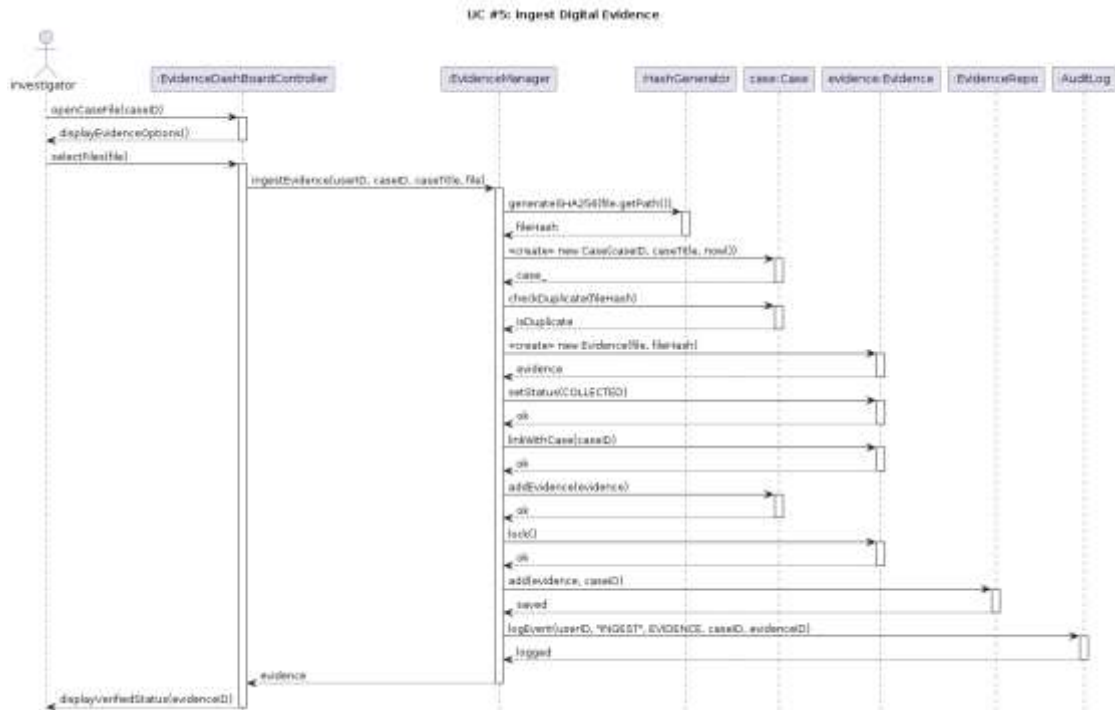
3. Initiate Investigation



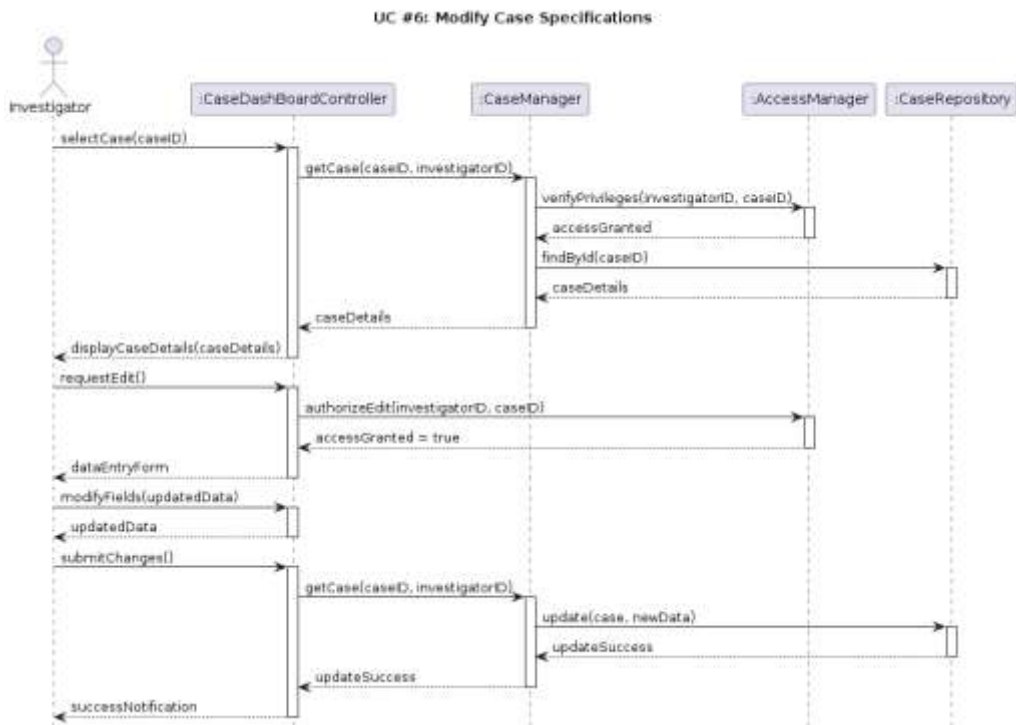
4. Authorize Case Access



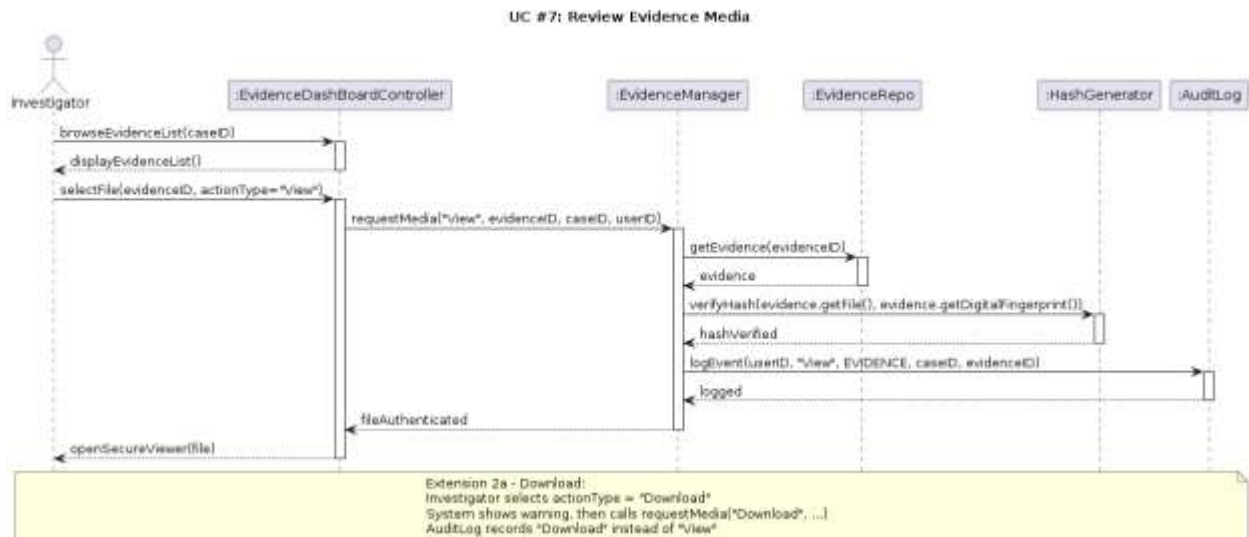
5. Ingest Digital Evidence



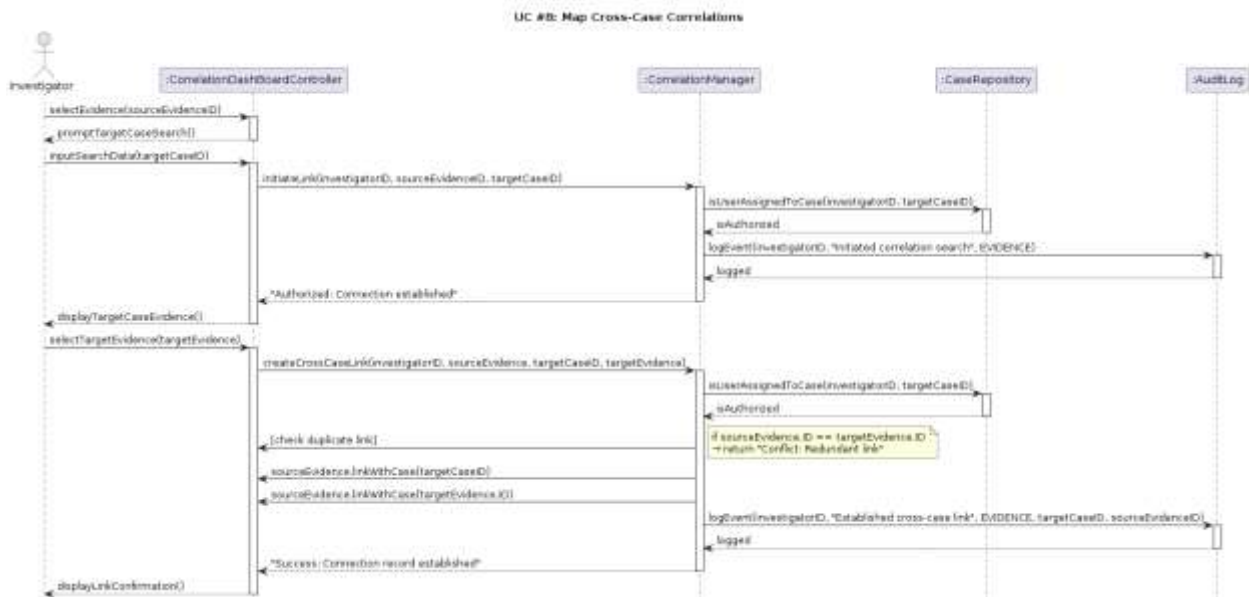
6. Modify Case Specifications



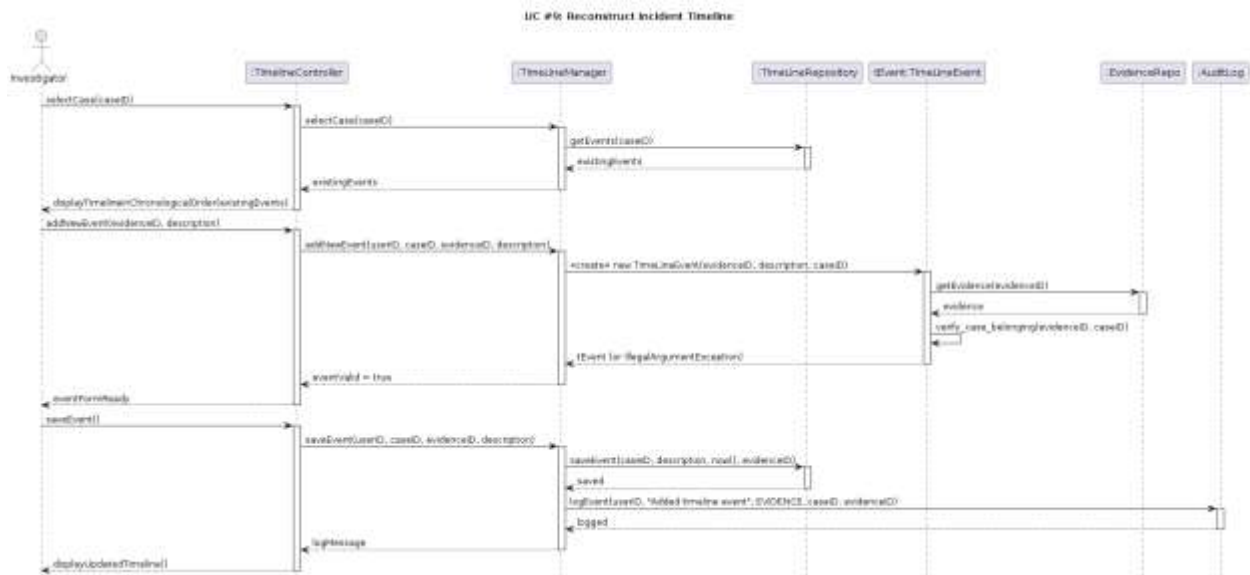
7. Review Evidence Media



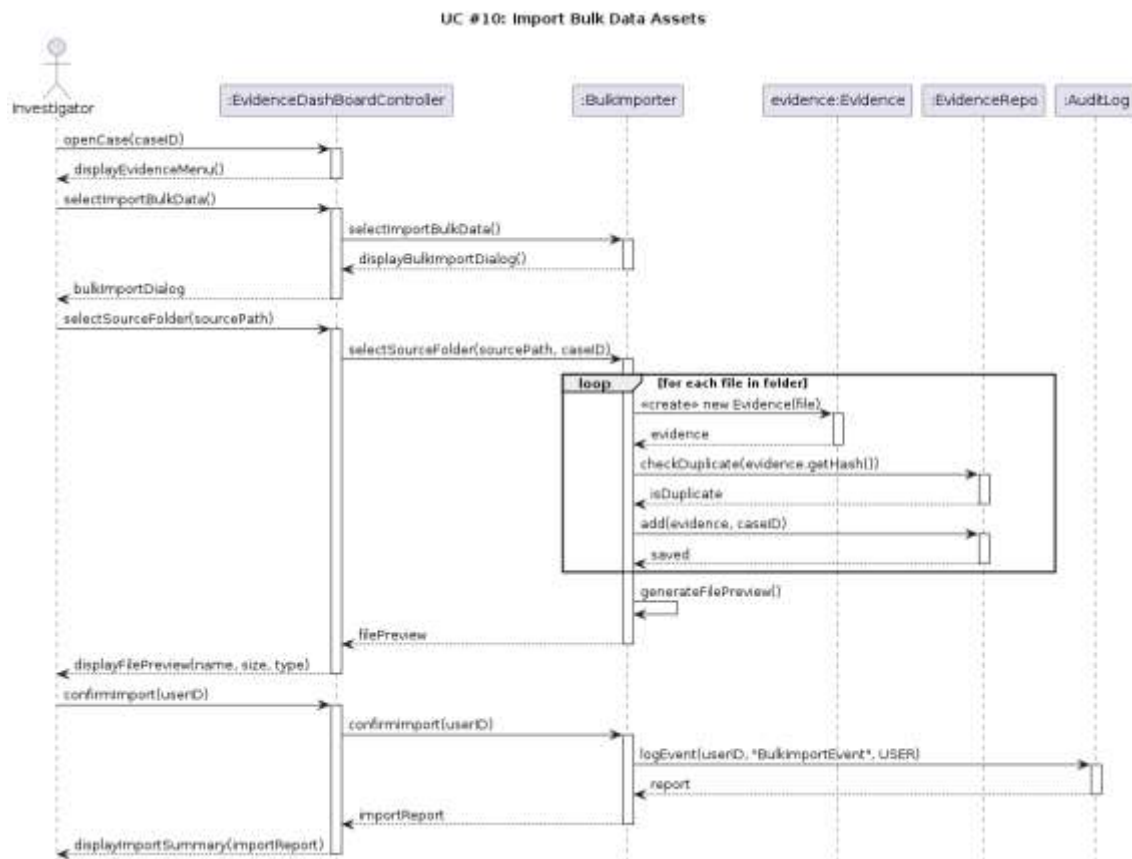
8. Map Cross – Case Correlations



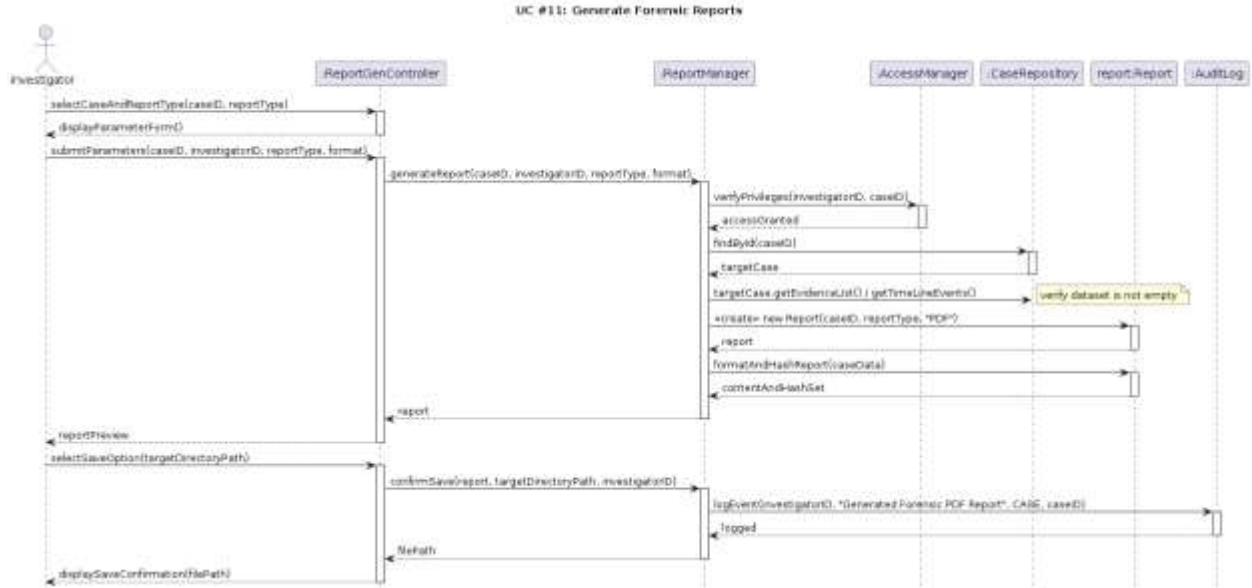
9. Reconstruct Timeline



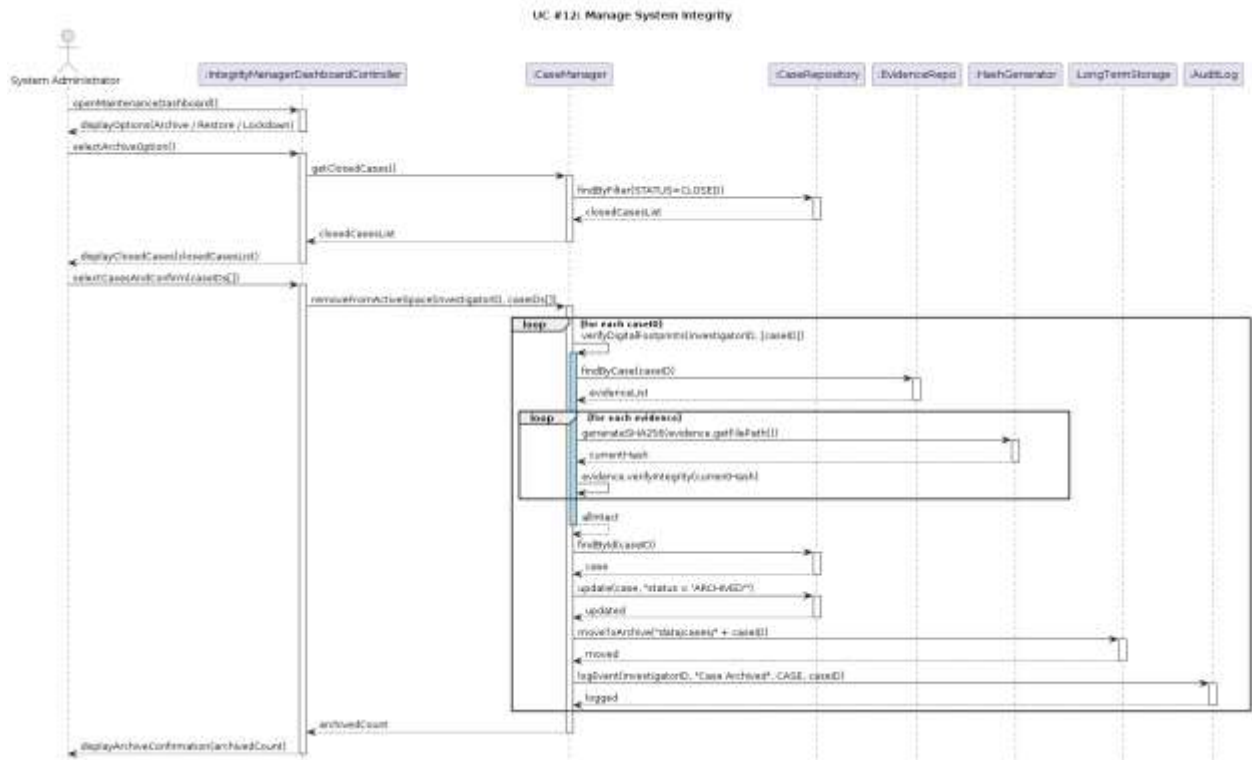
10.Import Bulk Data Assets



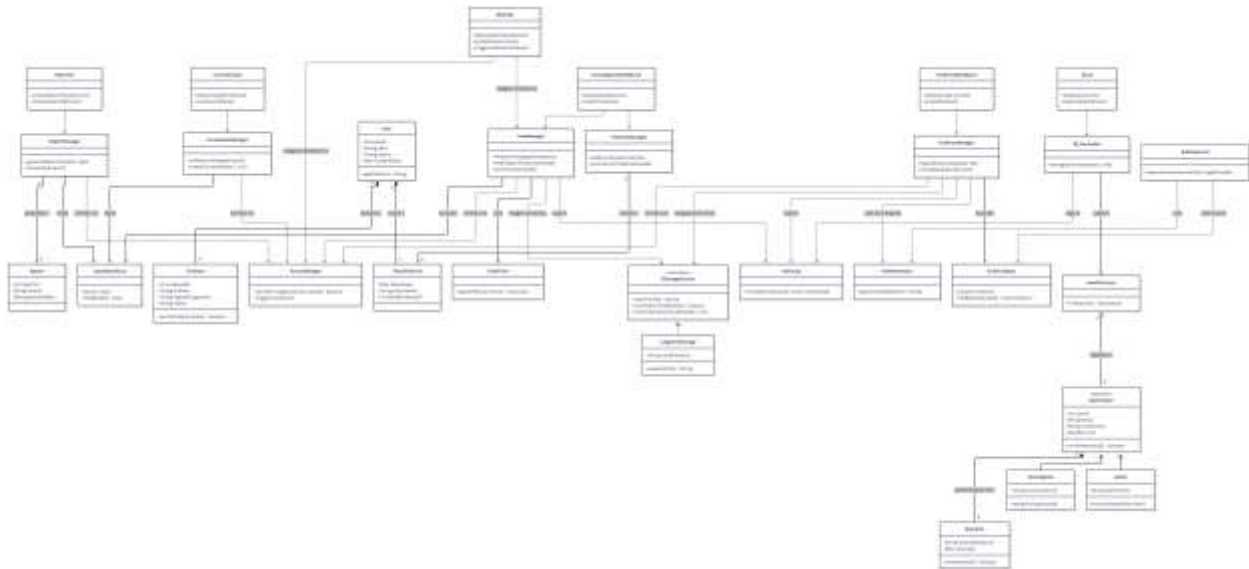
11. Generate Forensic Reports



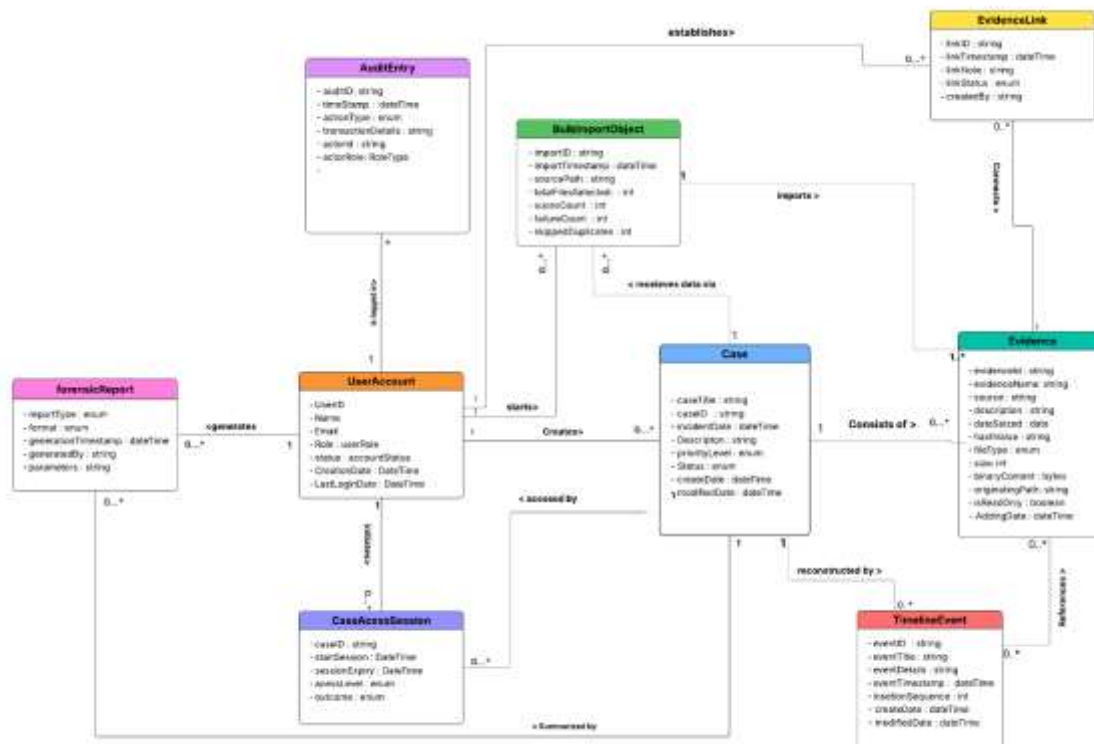
12. Manage System Integrity



3.3. Class Diagram

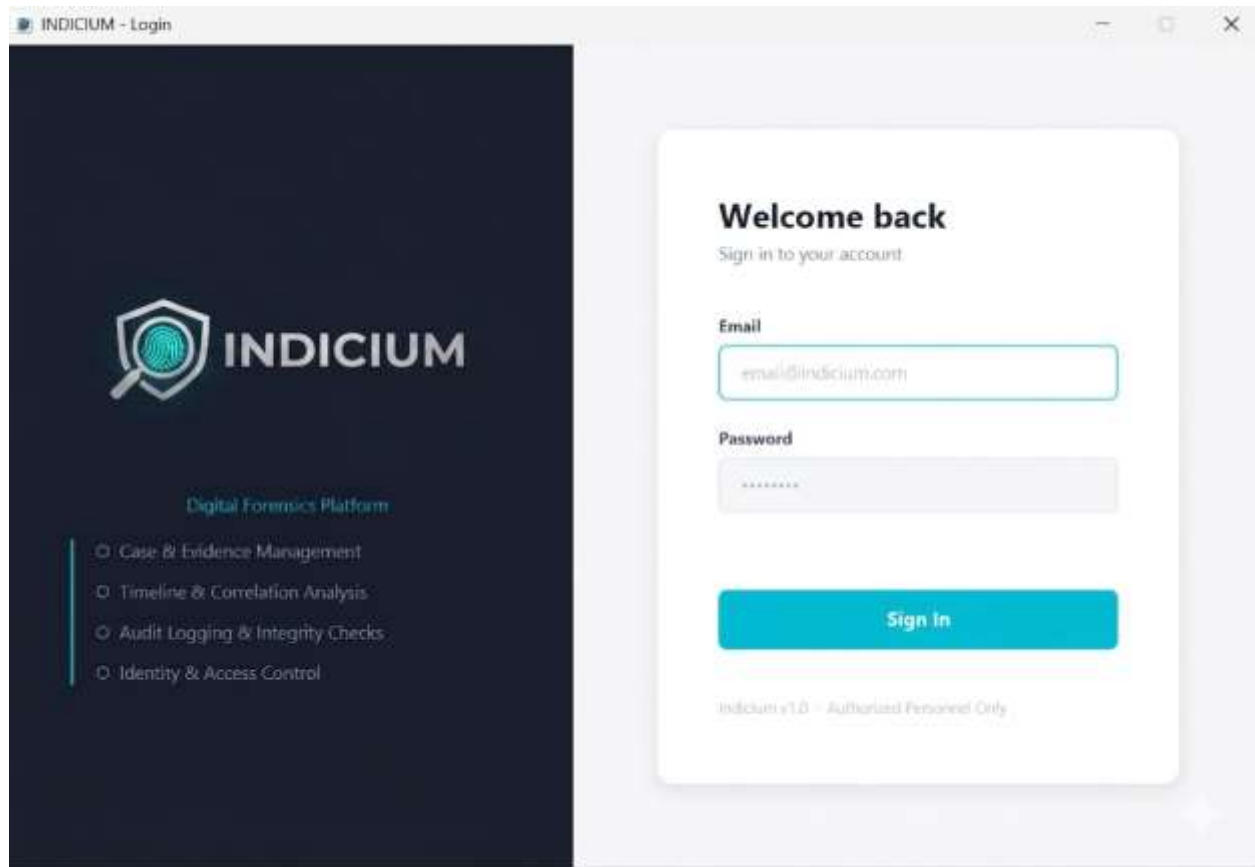


3.4. Domain Model



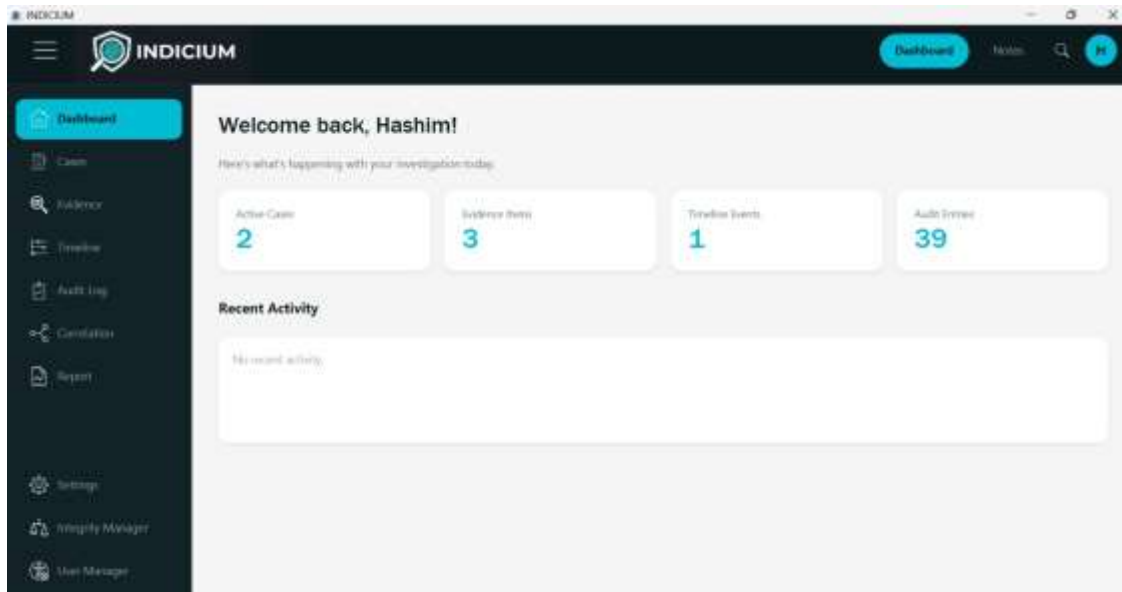
4. User Interface

4.1. Login Screen



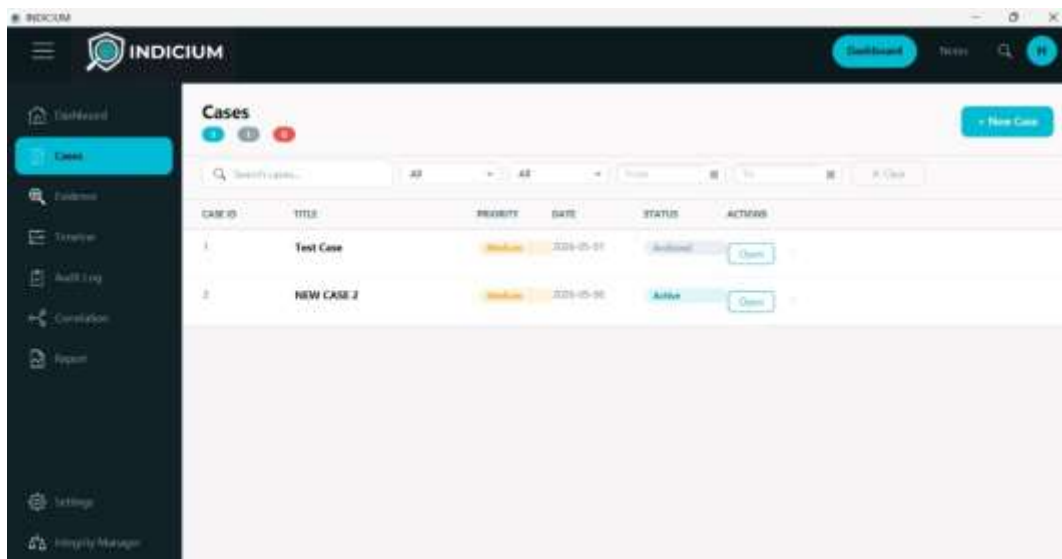
Users must log in using authorized Admin or Investigator credentials to access the system securely. Hence, each user is either the admin or an investigator. The login process ensures that only verified personnel can enter the platform and manage sensitive forensic data. Each user is required to provide a valid email and password for authentication before gaining access to system features and case records.

4.2. Dashboard Screen



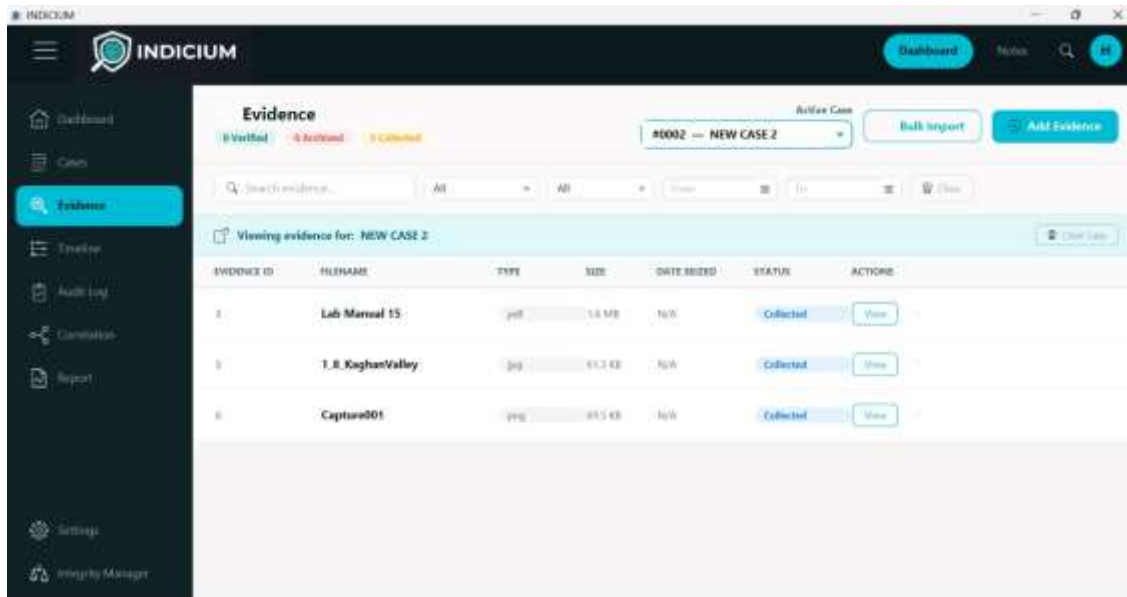
This is the main overview screen of the system. It provides an immediate summary of all the data currently stored in the database. It shows the total count of active cases, the number of files collected, and how many events have been recorded. It allows the team to see a quick snapshot of the entire system's workload in one place so they can monitor progress without searching through individual folders.

4.3. Cases Screen



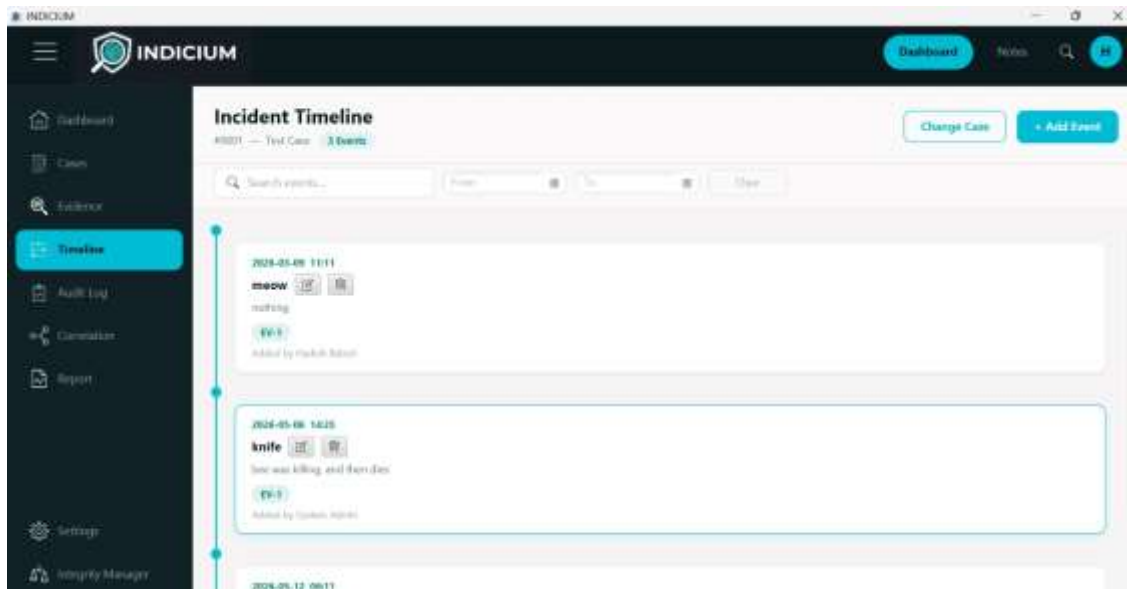
This screen is the central management area for all investigations. It shows a list of every project the team is currently working on, including details like the case title, how urgent it is, and whether it is still active or has been finished. It allows users to create new cases, open existing ones to see more details, and change the status of a project to keep the workspace organized.

4.4. Evidence Screen



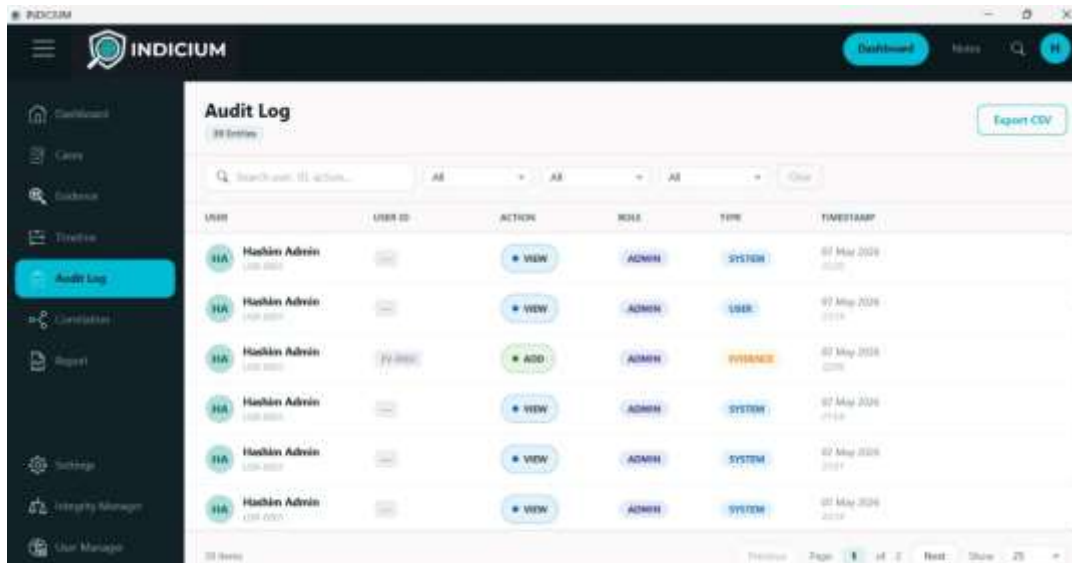
This is the storage and management area for all digital files. It shows a detailed list of every piece of proof gathered for a specific case, including the file name, its size, and what type of file it is (like a photo or a document). It allows the team to upload new files, view the contents of saved evidence, and track exactly which items have been collected and verified for the investigation.

4.5. Timeline Screen



This screen is used for chronological mapping and storytelling. It shows a vertical list of events arranged by the date and time they occurred. Each entry includes a description of the event and links it to specific pieces of evidence. It allows the team to visualize the sequence of an incident from start to finish, making it easier to understand how a situation developed over time.

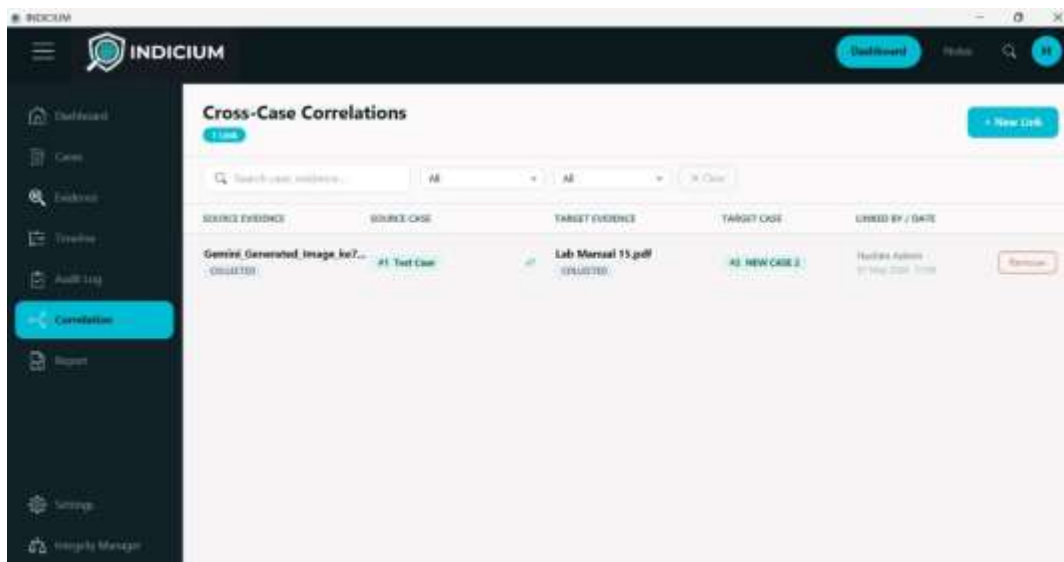
4.6. Audit Log Screen



USER	USER ID	ACTION	ROLE	TYPE	TIMESTAMP
Hachin Admin	1000000	VIEW	ADMIN	SYSTEM	07 May 2026 00:00
Hachin Admin	1000000	VIEW	ADMIN	USER	07 May 2026 00:00
Hachin Admin	1000000	ADD	ADMIN	EVIDENCE	07 May 2026 00:00
Hachin Admin	1000000	VIEW	ADMIN	SYSTEM	07 May 2026 00:00
Hachin Admin	1000000	VIEW	ADMIN	SYSTEM	07 May 2026 00:00
Hachin Admin	1000000	VIEW	ADMIN	SYSTEM	07 May 2026 00:00

This is the security and tracking module of the system. It shows a permanent, chronological list of every single action taken by any user within the software. It provides information on who performed a task, what they did, and the exact time it happened. It allows for total accountability by ensuring that no action goes unrecorded, which protects the integrity of the data.

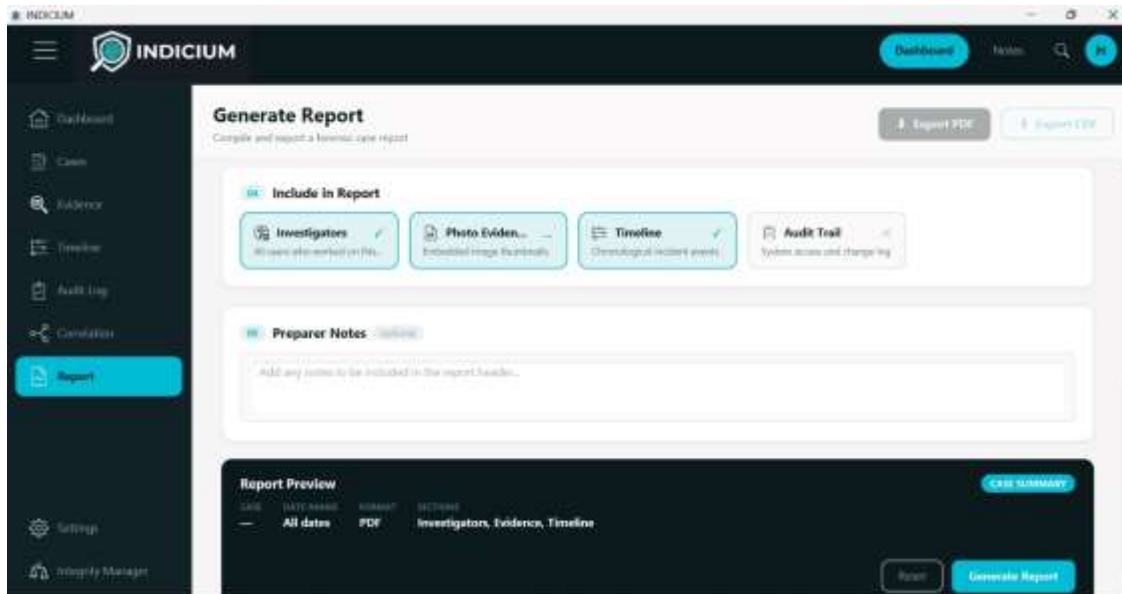
4.7. Correlation Screen



SOURCE EVIDENCE	SOURCE CASE	TARGET EVIDENCE	TARGET CASE	LINKED BY / DATE
Genital Generated Image ke7... (COLLECTED)	A1 Test Case	Lab Manual 15.pdf (COLLECTED)	A2 NEW CASE 2	Hachin Admin - 07 May 2026 10:00

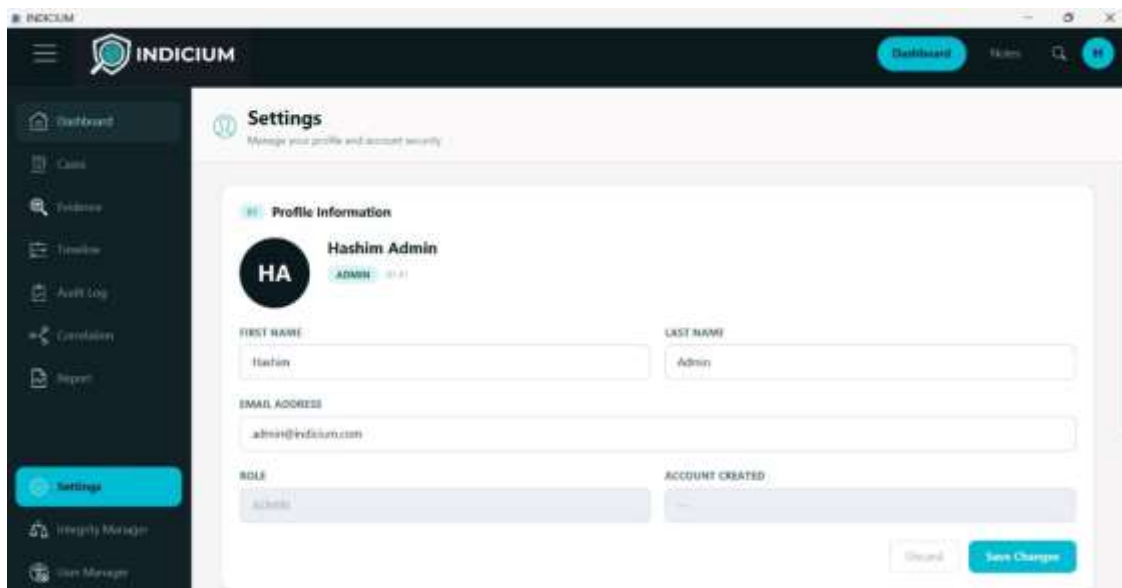
This is the advanced analysis tool of the platform. It shows connections between different pieces of evidence that belong to separate cases. It provides a way to link a file in "Case A" to a related file in "Case B." It allows the team to identify patterns and overlaps across the entire system, revealing if the same information is appearing in multiple investigations.

4.8. Report Screen



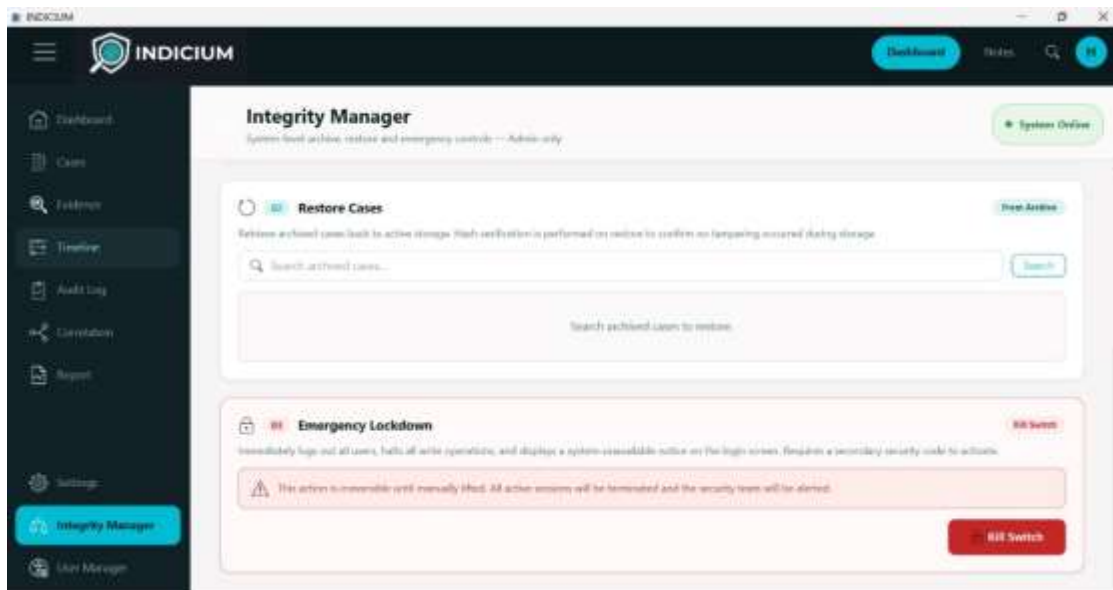
This is the final output and document generation area. It shows various options for what information can be included in a final summary, such as the list of investigators, the evidence photos, and the timeline of events. It allows the team to add their own custom notes and then automatically creates a professional document that can be exported as a file to be shared with others.

4.9. Settings Screen



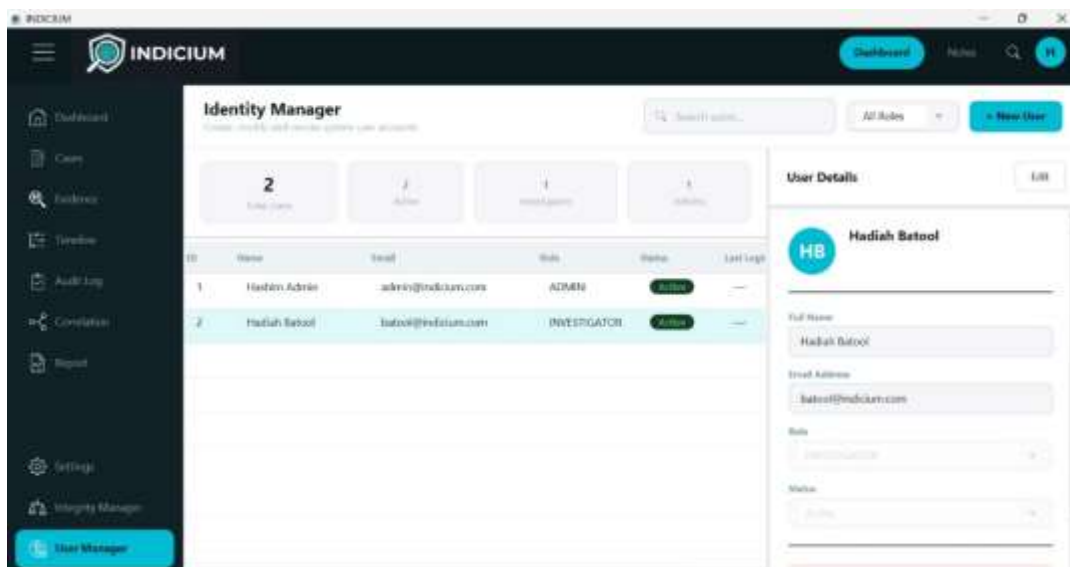
This is the personal profile and security management area. It shows the individual user's information, such as their full name, email address, and assigned role within the project. It allows the user to update their personal details and manage their account security to ensure their digital identity remains accurate and protected while using the system.

4.10. Integrity Manager Screen



This is the administrative control center for system safety. It shows options for recovering archived data and provides a high-level view of the system's online status. It allows an administrator to perform hash verifications to ensure files haven't been tampered with and includes an "Emergency Lockdown" feature to immediately secure all data and log out all users if a security threat is detected.

4.11. User Manager Screen



This screen handles the identity and access levels of everyone using the platform. It shows a list of all registered accounts, their email addresses, their specific roles (like Admin or Investigator), and their current status. It allows the team to create new user profiles, modify existing account details, and revoke access to ensure only authorized people can enter the system.

